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Household Portfolio Composition and Heterogeneous Monetary Policy Transmission in the Euro Area

by

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Abstract

This paper examines how household portfolio composition, beyond the level of wealth, shapes the transmission of monetary policy to consumption in the euro area. Using data from the ECB Household Finance and Consumption Survey for ten countries, we construct three measures of portfolio exposure: a liquidity index, a housing asset share and an equity asset share. We estimate with local projections how each governs the consumption response to high-frequency monetary policy shocks. The three channels transmit in the order of the assets' payoff horizon. Liquidity cushions the response on impact, housing deepens the contraction over the first quarters and equity matters only from the second year and only after strong prior returns. A four agent New Keynesian model calibrated to German data reproduces the same ranking of responses, through redistribution rather than consumption smoothing. Transmission is heterogeneous in the composition of wealth, and the channels line up by the payoff horizon of the assets that carry them.

Résumé

Cette étude examine comment la composition du portefeuille des ménages, au-delà du seul niveau de patrimoine, façonne la transmission de la politique monétaire à la consommation dans la zone euro. À partir des données de l'enquête sur les finances et la consommation des ménages (HFCS) de la BCE pour dix pays, nous construisons trois mesures d'exposition du portefeuille: un indice de liquidité, la part de l'actif immobilier et la part de l'actif en actions. Nous estimons, à l'aide de projections locales, la manière dont chacune régit la réponse de la consommation aux chocs de politique monétaire à haute fréquence. Les trois canaux se transmettent selon l'horizon de rendement des actifs. La liquidité amortit la réponse à l'impact, l'immobilier accentue la contraction au cours des premiers trimestres et les actions n'interviennent qu'à partir de la deuxième année, seulement après une forte hausse antérieure des cours boursiers. Un modèle néo-keynésien à quatre agents, calibré sur les données allemandes, reproduit ce même ordre, par le biais de la redistribution plutôt que du lissage de la consommation. La transmission est hétérogène selon la composition du patrimoine et les canaux s'ordonnent selon l'horizon de rendement de l'actif dont ils dépendent.

Keywords: Monetary policy transmission; Household heterogeneity; Portfolio composition; Collateral constraints; State dependence; Local projections; Four agent New Keynesian model; Household Finance and Consumption Survey; Euro area

JEL Classification: E52, D31, E21

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This paper uses data from the Eurosystem Household Finance and Consumption Survey. The results published and the related observations and analysis may not correspond to results or analysis of the data producers.

The work is the responsibility of the author, in no way does it engage the responsibility of the University, nor of the supervising Professor.

1 INTRODUCTION

Households differ not only in how much wealth they hold but in what that wealth consists of. In Germany, real estate accounts for 48% of household gross assets and in Italy 71%. The two countries share a currency, a central bank and a policy rate, yet their households enter a monetary policy shock holding different assets, and the same rate change does not reach a deposit and a mortgaged house in the same way. Since a shock reaches a household through the income it earns, the value of what it holds and the credit it can obtain against it, the distribution of wealth alone does not fully describe the transmission of monetary policy, as the composition of household portfolios also matters. We order these assets by their payoff horizon, that is, by how far in the future lie the returns the policy rate reprices. A deposit pays off now, a house delivers services and capital gains over decades, and equity is a claim on a dividend stream with no end date. The ordering matters because the horizon governs how the rate reaches each household. Short horizon payoffs are repriced as flows, so the shock arrives through income, while distant payoffs are discounted to a price today, so the shock arrives through that price. The mechanism therefore shifts as we move along the axis, from a flow channel at the short end to two valuation channels further out.

Three channels follow this order. At the shortest horizon, the cash-flow channel moves the debt service of a borrower and the interest income of a saver, both repriced on impact. We capture it through the liquidity of a household's assets, since a household whose wealth converts easily into spending can better absorb a change in cash flow than one whose wealth is illiquid. At intermediate horizon, the collateral and housing channel runs through both the value of the house and the credit obtainable against it, and we capture it through the share of housing in gross assets. At the longest horizon, the valuation channel runs primarily through price, and we capture it through the share of listed equity. Payoff horizon and liquidity are related but not the same, equity pays off furthest in the future and is also the easiest of the three to sell, whereas a house is both long-lived and slow to sell.

A large literature establishes that household heterogeneity shapes monetary transmission. In HANK models, heterogeneity in marginal propensities to consume and in households' access to liquid wealth changes the relative importance of direct and indirect transmission channels (Kaplan, Moll, et al. 2018), while the composition of household portfolios between assets with

different degrees of liquidity further affects transmission (Luetticke 2021). Euro area evidence either decomposes the consumption response into balance-sheet channels (Slacalek et al. 2020) or relates national responses to the prevalence of constrained households (Almgren et al. 2022). We estimate the differential response of three portfolio positions within a single specification, and ask whether each depends on the state of the asset market through which it runs.

This paper therefore asks three questions. Do households, whose portfolios differ along this ordering, display measurably different consumption responses to a common euro area monetary policy shock? Does the strength of a channel depend on the state of the asset market it runs through? Is the housing channel stronger after house prices have been rising, and the equity channel stronger after equity prices have? And can a tractable structural model, with one agent per position, reproduce the ordering the data delivers?

Household portfolios come from Wave 4 of the ECB Household Finance and Consumption Survey (European Central Bank 2022), covering ten euro area countries and 55,638 households, from which we build three exposure measures: a graded liquidity index, a housing asset share and an equity asset share. The outcome is quarterly real private consumption, and the identifying variation comes from the high-frequency monetary policy surprises of Jarociński and Karadi (2020). We estimate the responses by local projections (Jordà 2005), interacting the common shock with each exposure measure and, for housing and equity, with the prior state of the corresponding asset market. We then build a four agent New Keynesian model, one type per position, calibrated to German HFCS data. Its purpose is not to replicate the exact empirical magnitudes but to ask which mechanism produces the ordering the projections find.

Three results follow. The liquidity channel is present, peaks early and is transitory: comparing a relatively illiquid portfolio with a relatively liquid one offsets about 0.45 percentage points (pp) of the cumulative contraction at $h = 1$, roughly 11% of the average response, and the differential fades within six quarters. The housing channel operates in the same early window, but briefly and with the opposing sign, with housing-rich cells contracting by about 0.33 pp more within the first quarters before the difference reverses. The equity channel is weak on average but strongly state-dependent, the unconditional differential is imprecisely estimated, but at medium horizon a tightening arriving after a period of strong equity returns produces a substantially deeper cut among high-equity households. No such state dependence appears in the housing market. The model reproduces the same ordering, but reveals that a similar sign can hide different

mechanisms. The collateral-constrained homeowner contracts most and the bond saver least, with the hand-to-mouth agent and the shareholder in between. Yet the saver is cushioned not because it draws down liquid assets, but because the unexpected deflation raises the real value of its nominal claim and the borrower's deleveraging arrives as repayment, two transfers exactly mirrored in the borrower's budget, which cancel in the aggregate, so the saver's advantage is redistribution not smoothing. A regression on portfolio composition cannot separate mechanisms with a shared sign. The model can and it generates the same ordering with no smoothing at work, so the empirical sign does not by itself identify the mechanism behind it.

The rest of the paper is structured as follows. Section 2 reviews the literature on heterogeneous transmission. Section 3 describes the data and the construction of the exposure measures, and Section 4 their distribution. Section 5 sets out the local projections framework, and Section 6 reports the results. Section 7 develops and calibrates the four agent model and confronts it with the empirical results. Section 8 concludes.

2 LITERATURE REVIEW

2.1 From Representative-Agent to Heterogeneous-Agent New Keynesian Models

In the canonical representative-agent New Keynesian (RANK) model, the household-demand side of monetary transmission is organised around the consumption Euler equation. A change in the policy rate changes the expected real interest rate and therefore the intertemporal relative price of consumption (Clarida et al. 1999). With a single unconstrained household, transmission is by construction independent of the distribution of wealth and portfolios, an assumption that overlooks the high marginal propensities to consume associated with low liquid wealth (Kaplan, Violante, and Weidner 2014). Building on the rule-of-thumb consumers of Campbell and Mankiw (1989), two-agent New Keynesian (TANK) models reintroduce heterogeneity. Bilbiie (2008) shows that limited asset-market participation can substantially alter the aggregate response to interest-rate policy and Debortoli and Galí (2025) demonstrate that a suitably specified TANK approximates well far richer economies. These tractable frameworks, however, simplify household balance sheets substantially. Heterogeneous-agent New Keynesian (HANK) models relax this assumption at the cost of substantial analytical and computational complexity. In Kaplan, Moll, et al. (2018), a liquid–illiquid asset structure generates wealthy hand-to-mouth households, and indirect general-equilibrium effects operating through labour income dominate the direct consumption response, yet the model nevertheless contains a single illiquid asset, setting aside the distinction between collateral and valuation channels. Luetticke (2021) shows that wealth composition shapes transmission but housing and collateral remain absent. Auclert (2019) studies monetary transmission through three redistributive channels: an earnings heterogeneity channel, a Fisher channel, and an interest-rate exposure channel. We instead distinguish three portfolio-related mechanisms: cash flow, housing and collateral, and equity valuation, which are examined empirically and represented by distinct household types in the structural model.

2.2 Empirical Evidence on Balance Sheet Heterogeneity

For the euro area, Slacalek et al. (2020) combine HFCS microdata with aggregate monetary policy responses to decompose the consumption response into a direct and an indirect channel, the latter of which dominates. Tzamourani (2021) computes UREs directly from the HFCS and Adam and Zhu (2016) documents the Fisher channel redistribution in the euro area, both highlighting substantial cross-country differences in household balance-sheet exposures, including differences associated with debt and mortgage positions. Closest to our design are Almgren et al. (2022), who relate country-specific LP-IV responses to HtM shares, and Akarsu et al. (2026), who interact a time-invariant HtM share with the monetary policy shock in a panel LP. Existing studies typically examine balance-sheet channels separately. By studying liquidity, housing, and equity exposures within the same empirical framework, we can compare the relative importance of these channels.

2.3 The Cash-Flow Channel and Portfolio Liquidity

The liquidity channel builds on the wealthy-hand-to-mouth mechanism where households can have substantial net worth yet high MPCs when their wealth is held primarily in costly-to-access illiquid assets (Kaplan and Violante 2014). This mechanism motivates our liquidity index. Di Maggio, Kermani, Keys, et al. (2017) identifies the cash-flow channel through the repricing of U.S. adjustable-rate mortgages, capturing only the liability side and indebted homeowners. Our index instead measures how easily households can draw on their assets to smooth consumption after a shock, a distinction represented by agents 1 and 2 in the model.

2.4 The Housing and Collateral Channel

Kiyotaki and Moore (1997) show how collateral values and borrowing capacity interact, generating a feedback mechanism between asset prices and credit constraints. Cloyne et al. (2020) find that mortgagors in the United Kingdom and United States adjust spending strongly to interest-rate changes, with balance sheet constraints and income effects amplifying the response. Pica (2026) shows that transmission is stronger where adjustable-rate mortgages and homeownership are more prevalent. While Pica focuses on housing and mortgage institutions, our exposure measure

captures the household's housing position within the portfolio. Agent 3 represents this channel in the model.

2.5 The Equity and Valuation Channel

Monetary policy can affect household consumption through changes in equity values. Bernanke and Kuttner (2005) show that unexpected monetary policy changes affect stock prices mainly through revisions to expected excess equity returns. Using Swedish administrative data, Di Maggio, Kermani, and Majlesi (2020) show that households do increase consumption after stock-market gains, but the response is much smaller among wealthier households. This is particularly relevant because equity holdings are concentrated at the top of the wealth distribution. We additionally ask whether this valuation channel depends on recent stock-market conditions. In the structural model, this channel is represented by Agent 4, the equity holder.

3 DATA AND EXPOSURE MEASURES

The empirical analysis combines three structurally distinct data sources: cross-sectional household microdata from the Eurosystem Household Finance and Consumption Survey (HFCS) by the European Central Bank (2022), quarterly aggregate private consumption series from the ECB Data Portal (2026c), and monetary policy shocks identified from the high-frequency surprises of Jarociński and Karadi (2020). The HFCS provides cross-sectional variation in household portfolio composition across countries and the wealth distribution, which we summarise below into three exposure measures. The macro series provides the time-series outcome and the shock the identifying variation. These three sources are combined into a country × decile × quarter panel. The shock varies over time only, the exposure measure varies across countries and wealth deciles and the consumption outcome varies across countries and over time. Because consumption is published as a national aggregate, the outcome is constant across the deciles of a given country and quarter. The decile dimension refines the measurement of portfolio exposure without adding variation in the outcome.

3.1 Data Sources

3.1.1 Household Finance and Consumption Survey

The HFCS provides a cross-sectional snapshot of household balance sheets rather than a household panel. The analysis uses Wave 4 (2021), which collects detailed household-level information on demographics, labour-market status, income, consumption, and the composition of assets and liabilities. Fieldwork for this wave was conducted between the first half of 2020 and the first half of 2022 (European Central Bank 2023). The analysis focuses on ten euro area countries: Austria, Belgium, Germany, Spain, Finland, France, Ireland, Italy, Netherlands and Portugal, which together account for a major share of euro area economic activity. Portfolio exposures are observed at the survey date and are therefore treated as time-invariant characteristics in the empirical analysis. Heterogeneity in monetary policy transmission is identified by interacting these exposures with time-varying monetary policy shocks.

An important aspect of the HFCS is that it handles item non-response through multiple imputation.

For each missing value, five values are drawn by stochastic imputation conditional on observed household characteristics, producing five complete versions of the dataset called implicates. Multiple imputation propagates the uncertainty associated with missing information into the standard errors and supports more efficient point estimation (Rubin 1987, chap. 3; European Central Bank 2023). We therefore estimate each specification separately on each of the five implicates and combine the results using the rule of Rubin (1987)[1].

We use the HFCS cross-sectional household weight `HW0010` to construct population-representative exposure measures [2]. The weight is based on the inverse probability of household selection and is adjusted for differential unit non-response and calibrated to external population totals. It therefore determines how strongly each surveyed household contributes to population-level estimates. The sum of the weights across sampled households in a country is an estimate of the total number of households in that country, as reported in Table 3.1. Table 3.2 describes the key variables used in the analysis.

Table 3.1: **HFCS Wave 4 sample composition.** By country.

Country	**ISO**	**Unweighted N**	**Repr. HH (M)**
Austria	AT	2,293	4.07
Belgium	BE	2,130	5.02
Finland	FI	9,474	2.79
France	FR	10,253	30.27
Germany	DE	4,119	40.86
Ireland	IE	6,020	1.90
Italy	IT	6,239	25.32
Netherlands	NL	2,690	8.04
Portugal	PT	6,107	4.16
Spain	ES	6,313	18.82
Total	$C = 10$	**55,638**	**141.25**

Notes. Unweighted counts are from implicate 1. Representative households are in millions, derived from `HW0010`, and averaged across $M = 5$ implicates.

[1] Point estimates are averaged across implicates, and the total variance is $T = W + \left(1 + \frac{1}{M}\right) B$, where W is the average Driscoll–Kraay within-implicate variance, B is the variance of estimates across implicates, and $M = 5$.

[2] The HFCS also provides 1,000 replicate weights (`WR0001`–`WR1000`) for design-based variance estimation. These are not used here. Reported standard errors instead combine Driscoll–Kraay within-implicate uncertainty with Rubin's rules across implicates and therefore do not separately account for sampling uncertainty in the constructed HFCS exposure measures.

Table 3.2: **HFCS Wave 4 key variables.** Used in the analysis.

Variable	Label	Role in the analysis
DN3001	Net wealth	Ranks households into within-country net-wealth deciles.
DA3001	Total assets excl. public and occupational pension plans	Denominator for housing and equity portfolio shares.
DA2105	Shares, publicly traded	Numerator of the equity share exposure; captures direct exposure to equity price movements.
DA1110	Value of household's main residence	Main component of housing wealth and of the housing-share exposure.
DA1120	Other real estate	Added to HMR value to measure total real-estate exposure.

Notes. The full set of asset components used in the liquidity index is reported separately in Table 3.4.

3.1.2 Aggregate Consumption

The consumption series is the quarter-on-quarter growth rate of real private final consumption expenditure by country, drawn from the ECB Data Portal (2026c) [3]. The outcome is constructed as the cumulative log change in consumption relative to the quarter preceding the monetary policy shock, so that the estimated response reads directly as a cumulative percentage change relative to the pre-shock level. This formulation is preferred to HP-filtered levels because it avoids the choice of a smoothing parameter and end-of-sample distortions (Hamilton 2018). The series is published as a national aggregate, so the outcome is observed at country × quarter frequency only.

3.1.3 Monetary Policy Shocks

Monetary policy shocks are identified following Jarociński and Karadi (2020), from the high-frequency co-movement of interest-rate and equity price surprises in a narrow window around each ECB Governing Council announcement. The underlying surprises, changes in short-term OIS interest rates and in the EURO STOXX 50, are taken from the Euro Area Monetary Policy Event-Study Database (Altavilla et al. 2019). Two structurally distinct types of news move markets at a given announcement. A pure monetary policy shock moves interest rates and equity prices in opposite directions, whereas a central bank information shock moves them together, by

[3]The ECB series reports the quarter-on-quarter percentage change in the chain-linked volume measure of private final consumption expenditure (reference year 2020 = 100), calendar and seasonally adjusted. We convert this percentage change into log growth before estimation.

revealing the ECB's assessment of the outlook. Jarociński and Karadi (2020) separate the two by imposing sign restrictions on this co-movement and reporting the median of the identified set. We use the median series [4] `MP_median` as the baseline shock [5]. The shocks, provided at monthly frequency and set to zero in months without a Governing Council meeting, are aggregated to quarterly frequency by summation and rescaled so that one unit corresponds to a 25 basis point (bp) surprise, with a positive value denoting an unexpected tightening.[6]

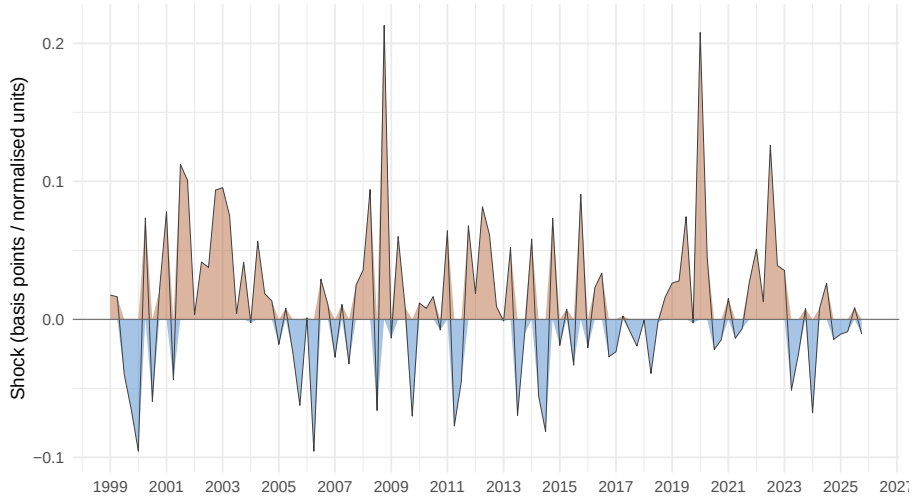


Figure 3.1: **ECB monetary policy shocks.** Quarterly aggregated MP shock (median) by Jarociński and Karadi (2020).

Notes. Orange indicates tightening; blue indicates easing.

3.1.4 House and Equity Prices

The empirical analysis additionally uses a country-level residential property price index from the ECB Data Portal (2026a) [7], at quarterly frequency, with one index per country representing the whole national territory. Country-level equity prices are taken from the OECD (2026): national share price indices, one per country, expressed as quarterly indices.[8] The liquidity and equity specifications are estimated from 1999Q1, matching the consumption and shock data. The housing specification runs on a shorter panel, starting between 2006Q2 and 2011Q2 depending on the country, because the house price index begins later and at different dates across countries.

[4] We use the updated series maintained online by the authors (Jarociński and Karadi 2025)

[5] This measure is preferred to the poor man's sign restriction because it allows monetary policy and central bank information shocks to coexist within the same ECB announcement.

[6] The 25 bp unit is a scaling convention rather than a typical shock. Over 1999–2025 the quarterly surprise has a standard deviation of 5.4 bp and never exceeds 21.3 bp in absolute value

[7] The index is transaction-based, rebased to 2015 = 100, and is not seasonally or calendar adjusted. Therefore we compute year-on-year log growth rates, which remove the seasonal pattern.

[8] Rebased to 2015 = 100.

3.2 Construction of Portfolio Exposure Measures

Household heterogeneity is summarised by three portfolio exposure measures, each intended to capture a distinct channel through which monetary policy affects consumption: a liquidity index, capturing the cash-flow channel discussed in Section 6.1; a housing asset share, capturing the collateral/house price channel discussed in Section 6.2; and an equity asset share, capturing the valuation/equity price channel discussed in Section 6.3. These measures are defined and described in Table 3.3. The liquidity index assigns asset-specific liquidity weights ranging from 1.00 for deposits, which are immediately spendable, to 0.05 for self-employment business wealth, which is the least readily convertible asset category. The full set of liquidity weights is reported in Table 3.4. The index is inspired by the liquid/illiquid asset distinction and the ordering in Kaplan and Violante (2014) and Kuypers and Marx (2019), but extended into a graded portfolio-liquidity score.[9]

Households are sorted into within-country net-wealth deciles[10], and the exposure measure is aggregated within each country × decile cell using the weight `HW0010`, giving $E_{c,d}$.

[9]The weights ℓ_j capture the relative liquidity ordering of asset classes, how quickly and cheaply each can be converted into cash, rather than precise magnitudes.

[10]Vigintiles or percentiles sharpen the resolution of the exposure distribution but shrink each cell, making its household-weighted mean noisier. Given HFCS sample sizes, percentiles leave too few households per cell for reliable estimates, while vigintiles improve little on deciles; Vigintile partitions are reported as robustness checks in Appendix II.

Table 3.3: **Portfolio exposure measures.** Definitions and construction.

	Liquidity Index	**Housing Share**	**Equity Share**
Formula	$L_i = \frac{\sum_j \ell_j a_{ij}}{\sum_j a_{ij}}$	$H_i = \frac{\text{DA1110}_i + \text{DA1120}_i}{\text{DA3001}_i}$	$E_i = \frac{\text{DA2105}_i}{\text{DA3001}_i}$
Numerator	Liquidity-weighted sum of assets across 13 components, where a_{ij} is household i's holding of asset j and $\ell_j \in [0, 1]$ is its liquidity weight.	Sum of main residence value (DA1110) and other real estate (DA1120).	Market value of directly held publicly listed shares (DA2105).
Denominator	Total value of the 13 included asset components.	Total gross household assets (DA3001).	Total gross household assets (DA3001).
Range	$[0.05, 1]$	$[0, 1]$	$[0, 1]$
Interpretation	Higher values indicate a more liquid portfolio; a score of 1 means all assets are fully liquid.	Share of total wealth held in real estate; higher values indicate stronger exposure to housing wealth.	Share of total wealth held in listed equities; higher values indicate greater exposure to equity price movements.
Exclusions	DA2108 is excluded from both numerator and denominator due to heterogeneous sub-components.	Households with DA3001 ≤ 0 or missing are excluded.	Households with DA3001 ≤ 0 or missing are excluded; DA2105 = NA is set to 0.
Cell aggregation	Weighted mean of the household index: $E_{c,d} = \frac{\sum_i w_i L_i}{\sum_i w_i}$.	Ratio of weighted totals: $E_{c,d} = \frac{\sum_i w_i (\text{DA1110}_i + \text{DA1120}_i)}{\sum_i w_i \text{DA3001}_i}$.	Ratio of weighted totals: $E_{c,d} = \frac{\sum_i w_i \text{DA2105}_i}{\sum_i w_i \text{DA3001}_i}$.

Table 3.4: **Liquidity index.** Asset components and weights.

Variable	**Asset**	**Liquidity weight ℓ_j**
DA2101	Deposits (sight + savings)	1.00
DA2102	Mutual funds	0.85
DA2103	Bonds	0.80
DA2105	Publicly traded shares	0.70
DA2106	Managed accounts	0.60
DA2107	Money owed to household	0.40
DA1130	Vehicles	0.35
DA1131	Valuables	0.30
DA2104	Non-SE private business wealth	0.25
DA1110	Household main residence (HMR)	0.20
DA1120	Other real estate	0.15
DA2109	Voluntary pensions / whole-life ins.	0.10
DA1140	Self-employment business wealth	0.05
DA2108	Other assets (excluded)	—

4 DESCRIPTIVE STATISTICS

This section analyses the three portfolio-exposure measures across households and country × decile cells that enter the estimation.

Table 4.1: **Distribution of household portfolio-exposure measures**

Measure	Mean	Median	SD	P25	P75	Min	Max	n
Panel A: Household-level (weighted by HW0010)								
Liquidity index	0.407	0.304	0.249	0.225	0.504	0.050	1.000	55391
Housing share	0.499	0.651	0.398	0.000	0.869	0.000	1.000	55392
Equity share	0.010	0.000	0.057	0.000	0.000	0.000	1.000	55392
Panel B: Cell-level (country × decile)								
Liquidity index	0.416	0.320	0.171	0.288	0.542	0.246	0.818	100
Housing share	0.653	0.735	0.195	0.562	0.788	0.005	0.852	100
Equity share	0.008	0.005	0.012	0.002	0.010	0.000	0.092	100

Notes. Averaged across the 5 implicates. Panel A is the household-level distribution weighted by the survey weight HW0010. Panel B is the unweighted distribution across the country × wealth decile cells, the cross-sectional variation the local projections use.

Table 4.1 reports the distribution of the three exposure measures. The median and the 75th percentile of the equity exposure are both zero, so at least three quarters of the population hold no shares, while the mean of 0.010 shows that a thin upper tail exists. This anticipates a weak equity differential and a small population assigned to equity holdings. At least a quarter of households hold no real estate, so the 25th percentile of the housing share is zero, while the median household holds 65.1% of its gross assets in real estate. The standard deviation of 0.398 therefore overstates the dispersion among owners, because of the gap between owners and non-owners. The liquidity index is right-skewed, its mean is above the median. Panel B reports the same measures after aggregation. The standard deviation falls for all three and equity weakens the most, since most of household-level dispersion is internal to the cells.

Table 4.2 reports the country means that generate this variation. In eight of the ten countries real estate is the majority of gross assets, the exceptions being Germany and Austria. Shares never reach 2%, and the liquidity index ranges from about a third to just over a half of its scale. Equity exposure varies far less across cells than the other two measures, which is why the differential reported in Section 6.3 remains fragile. The final column reports median gross assets, since the liquidity index measures only the average liquidity of a portfolio, not its size, a household holding little wealth entirely in deposits scores at the top of the index which doesn't necessarily

Table 4.2: **Portfolio exposure by country.** HFCS Wave 4.

Country	ISO	Liquidity index	Housing share	Equity share	Median gross assets (€k)
Italy	IT	0.333	0.708	0.004	177
Spain	ES	0.349	0.749	0.007	166
Ireland	IE	0.363	0.735	0.006	266
France	FR	0.371	0.594	0.006	177
Portugal	PT	0.382	0.712	0.002	125
Belgium	BE	0.407	0.694	0.007	302
Finland	FI	0.445	0.694	0.019	156
Germany	DE	0.485	0.482	0.017	122
Netherlands	NL	0.506	0.704	0.007	232
Austria	AT	0.519	0.460	0.005	141
Cross-country mean		**0.416**	**0.653**	**0.008**	**186**

Notes. Country means of the three exposure measures, obtained by averaging the country × decile cell means over the ten deciles and over the five HFCS implicates. Countries are ordered by the liquidity index. Median gross assets is weighted by `HW0010`, averaged across implicates, and rounded to the nearest thousand euros. The liquidity index and median gross assets correlate at −0.17.

mean a higher buffer. Austria illustrates this, ranking highest on liquidity but among the lowest in median gross assets. The two correlate at −0.17 across the ten countries, so the index is not a proxy for portfolio size. We also check the correlation between the measures since they are built from overlapping components. The liquidity index and housing share correlate at −0.657 because the main residence is the largest component of most portfolios and enters the index as a low weight. More than half of their variation is nonetheless independent. The equity share is close to orthogonal to both, correlating at −0.14 and −0.13, so it carries information the other two do not.

5 EMPIRICAL STRATEGY

5.1 Local-Projection Framework

We estimate impulse response functions (IRFs) using the local projection (LP) method of Jordà (2005). Rather than iterating the estimated coefficients of a single vector autoregression (VAR) forward to trace out the dynamic response, the LP method runs a separate regression at each horizon h, projecting the cumulative outcome h quarters ahead directly onto the shock dated t and a set of controls. The LP approach offers two advantages in our setting. First, it does not require the full joint dynamics of the system to be specified. Each horizon is estimated by its own regression, so a misspecified or truncated lag structure is not carried forward across horizons. With sufficiently rich and unrestricted lag structures, LPs and VARs estimate the same impulse responses (Plagborg-Møller and Wolf 2021), the advantage is therefore not a different estimand but greater robustness to specification error in finite samples, at the cost of some efficiency at longer horizons. Second, the framework fits naturally with heterogeneity [11], the shock is interacted with a cross-sectional exposure measure that enters as an additional regressor, so differential transmission across cells can be estimated directly by OLS within one panel regression.

5.2 Baseline Specification, Controls, and Inference

$$y_{c,t+h} = \alpha_c + \beta_h s_t + \gamma_h \left(s_t \times \widetilde{E}^{l}_{c,d} \right) + \boldsymbol{\delta}_h' \mathbf{X}_{c,t} + \varepsilon_{c,d,t+h}, \qquad h = 0, \ldots, H, \tag{1}$$

where $y_{c,t+h}$ is the cumulative log growth of real private consumption in country c between $t-1$ and $t+h$. α_c denotes country fixed effects, which control for time-invariant differences across countries. s_t denotes the common euro area monetary policy shock in quarter t. The interaction term $s_t \times \widetilde{E}^{l}_{c,d}$ captures whether the response to a common monetary policy shock differs with the portfolio exposure of the country × decile cell[12]. In this baseline the exposure

[11] This exposure-interaction approach follows a growing literature that uses local projections to estimate heterogeneous responses to monetary policy across household and firm characteristics (Akarsu et al. 2026; Cloyne et al. 2020)

[12] The exposure measure is centered and scaled as $\widetilde{E}_{c,d} = \frac{E_{c,d} - \bar{E}}{0.10}$ where $\bar{E}$ is the cross country × decile mean exposure. Centering sets $\widetilde{E}_{c,d} = 0$ at average exposure, and $\widetilde{E}_{c,d} = 1$ denotes a cell whose exposure is 0.10 above the mean.

measure $\widetilde{E}^{l}_{c,d}$ is the liquidity index, the housing and equity specifications keep the same form and replace it with the housing asset share $\widetilde{E}^{ho}_{c,d}$ and the equity share $\widetilde{E}^{e}_{c,d}$. The coefficient β_h measures the average cumulative consumption response to a monetary policy shock at horizon h for a country × decile cell at the average exposure level. γ_h is the main coefficient of interest and measures the differential cumulative consumption response to a 25 bp tightening associated with a 10 pp higher liquidity index. The vector $\mathbf{X}_{c,t}$ contains a Covid dummy, lags of outcome, lags of shock and macroeconomic controls of real GDP growth, the change in the unemployment rate, and inflation (ECB Data Portal 2026b). All series entering the projection are tested for stationarity in Appendix IV.

This baseline specification assesses how the liquidity of a household's portfolio shapes its consumption response to monetary policy. A less liquid portfolio is expected to amplify the transmission of monetary policy to consumption (Kaplan and Violante 2014; Kaplan, Moll, et al. 2018). When interest rates rise, households whose wealth is concentrated in illiquid instruments cannot easily draw down assets to smooth spending. Therefore, they behave like wealthy hand-to-mouth households and display a higher marginal propensity to consume (Kaplan and Violante 2014; Cloyne et al. 2020). A monetary tightening raises the return to saving and the cost of borrowing, reducing consumption on average. The liquidity composition of the portfolio then governs how strongly each household's consumption reacts.

Lag orders are selected using an information-criterion grid search evaluated at horizon $h = 0$. The search is conducted separately for the liquidity, housing and equity specifications, using a fixed common sample. Across all three specifications within the considered grid, both AIC and BIC select the grid's upper boundary: 8 lags of outcome p_y, 6 lags of control p_X, and 4 of shock p_s. Additional details on the lag-selection procedure are reported in Appendix III.

The Covid-19 period poses a challenge: consumption swings that are far larger than in normal times can distort the regression and dominate the estimated impulse responses, as shown in Figure 5.1. However, dropping the period entirely would discard valuable observations and shorten the sample considerably. The baseline specification therefore adds a Covid dummy[13] equal to one whenever the horizon-h outcome window [t, t+h] overlaps 2020Q1–2021Q4, absorbing the average pandemic-window shift in cumulative consumption while retaining the rest of the sample.

[13] As an alternative the Covid dummy is replaced by year fixed effects reported as a robustness check in Appendix I.

The structure of the data, a cross-country panel in which all units are hit by the same ECB monetary policy shock, motivates the choice of standard errors. Heteroskedasticity-robust standard errors assume the errors are uncorrelated across observations, and country-clustered standard errors, while allowing arbitrary correlation within a country, assume independence across countries. Because the common shock and shared euro area disturbances make outcomes contemporaneously correlated across countries, both understate the true uncertainty. We therefore use Driscoll and Kraay (1998) standard errors, which are robust to general forms of cross-sectional and serial dependence as the time dimension T grows large (Hoechle 2007). The bandwidth is set to the maximum projection horizon to account for the overlapping-window autocorrelation induced by cumulative outcomes. This large-T, small-N regime matches the structure of the panel: 10 countries observed over a long quarterly sample. Because the shock is common to all units, the cross-section carries little independent information on the average response, so inference must rest primarily on the time dimension.

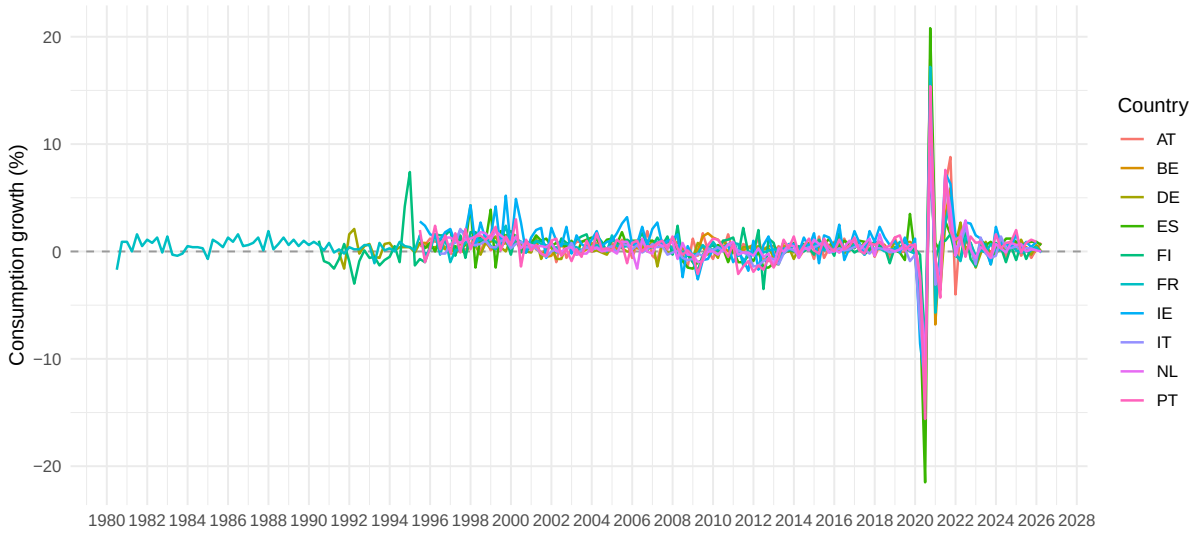


Figure 5.1: **Real private consumption growth.** By country, full sample.

6 EMPIRICAL RESULTS

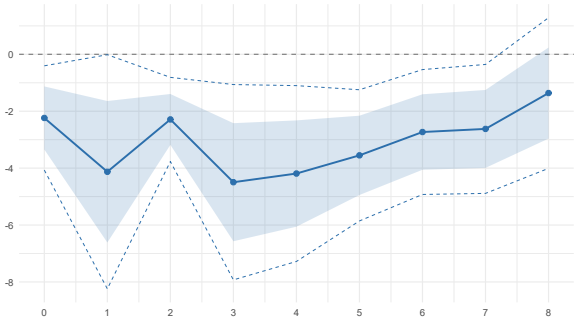
6.1 Liquidity Exposure and the Cash-Flow Channel

Figure 6.1a plots the cumulative log response of real private consumption to a 25 bp contractionary ECB monetary policy shock. All figures report 68% confidence intervals (approximately ±1 standard error) and 90% confidence intervals. The response is negative at every horizon, reaching a peak decline of approximately 4.5% at $h = 3$. From $h = 4$ onward it partially recovers, and by $h = 8$ the confidence band includes zero. The loss of precision at longer horizons is expected. The non-monotonicity around $h = 2$ is harder to interpret and may reflect residual pandemic-period dynamics that are not fully absorbed by the Covid-window dummy. The sign is the standard response of consumption to monetary policy. Higher policy rates raise the return to saving and the cost of borrowing, so households postpone consumption and reduce their current demand for goods and services.

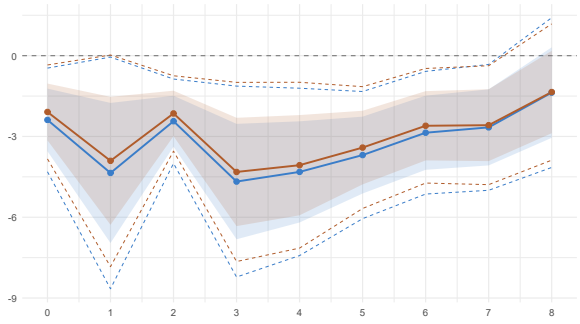
Figure 6.1b reports the cumulative response evaluated at the 25th and 75th percentile of liquidity exposure. Neither path reverses sign, both remain negative but differ in magnitude. The high-exposure curve lies above the low-exposure one at every horizon, indicating a dampening effect. Households with more liquid portfolios reduce consumption less after a monetary tightening.

Figure 6.1c provides the formal counterpart, plotting the differential response $\gamma_h \times \left(\widetilde{E}^{l}_{75} - \widetilde{E}^{l}_{25}\right)$ with its own confidence band. This is the appropriate object for heterogeneity, since the significance of the difference cannot be read off the overlap of the two separate confidence bands in Figure 6.1b. The differential is positive and statistically significant up to $h = 6$, before gradually converging to zero. Part of the consumption response is therefore attributable to differences in liquidity exposure. The effect peaks early and is transitory, which reflects the timing of the channel. At its peak ($h = 1$), moving from the 25th to the 75th percentile offsets approximately 0.45 pp of the cumulative contraction, corresponding to about 11% of the average response at that horizon. This magnitude is consistent with a cash-flow, or interest-rate-exposure, channel (Kaplan and Violante 2014; Auclert 2019). A monetary tightening weakens demand and tightens borrowing conditions. Households with illiquid portfolios cannot easily draw on their assets to smooth consumption, since a large share of their wealth is locked in housing, vehicles, pension assets, or other illiquid classes. When their cash flow weakens, they cut spending more

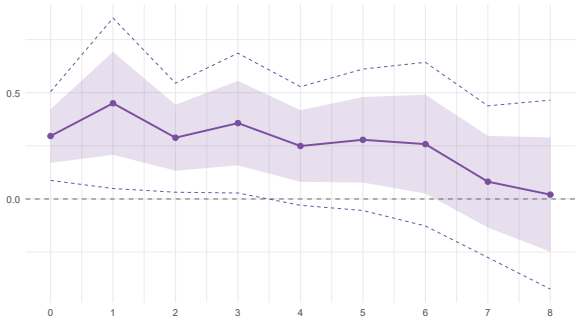
sharply. Households with more liquid portfolios can instead draw down deposits and bonds to maintain consumption and may partly benefit from the higher return on liquid safe assets after a tightening. Figure 6.1c captures this composition effect: holding the aggregate response fixed, cells with greater liquidity exposure contract by less.



(a) Average response



(b) By liquidity exposure



(c) Differential response

Figure 6.1: **Liquidity exposure and consumption responses to a monetary policy shock.**

Notes. In all panels the horizontal axis is the **horizon** h and the vertical axis is the **cumulative log change in real private consumption (pp)**. (a) reports the average cumulative response. (b) reports the response evaluated at low (blue, P25, $\widetilde{E}^{\,l} = -1.29$) versus high (orange, P75, $\widetilde{E}^{\,l} = 1.26$) liquidity exposure. (c) reports the differential response $\gamma_h \times (\widetilde{E}^{\,l}_{75} - \widetilde{E}^{\,l}_{25})$. Shaded band: 68% CI. Dashed lines: 90% CI.

Table 6.1: **Local projection estimates, liquidity exposure.** Transmission of a 25 bp contractionary shock to consumption, by liquidity-index exposure.

	β_h		γ_h	
h	Coef.	Std. err.	Coef.	Std. err.
0	−2.237*	1.114	0.116*	0.050
1	−4.128*	2.501	0.177*	0.096
2	−2.289*	0.899	0.113*	0.061
3	−4.495*	2.087	0.140*	0.078
4	−4.190*	1.878	0.098†	0.067
5	−3.551*	1.401	0.109†	0.079
6	−2.731*	1.334	0.101†	0.092
7	−2.622*	1.377	0.032	0.085
8	−1.361	1.609	0.008	0.106

Notes. * $|t| \geq 1.645$ (90%); † $|t| \geq 1$ (68%). Driscoll–Kraay SE. The fraction of missing information from multiple imputation is below 0.01 for all coefficients at every horizon.

6.2 Housing Exposure and the Collateral Channel

This specification extends the baseline in Equation (1) in two ways: the exposure measure $\widetilde{E}^{ho}_{c,d}$ is now the household housing asset share, and two house price terms are added to test whether the exposure channel depends on the state of the housing market:

$$y_{c,t+h} = \alpha_c + \beta_h s_t + \gamma_h \left(s_t \times \widetilde{E}^{ho}_{c,d}\right) + \rho_h \widetilde{HP}_{c,t-1} + \theta_h \left(s_t \times \widetilde{E}^{ho}_{c,d} \times \widetilde{HP}_{c,t-1}\right) + \boldsymbol{\delta}_h' \mathbf{X}_{c,t} + \varepsilon_{c,d,t+h}, \quad (2)$$

The variable $\widetilde{HP}_{c,t-1}$ is the lagged, centered, and scaled year-on-year growth of the house price index (ECB Data Portal 2026a) [14] The coefficient ρ_h captures the direct (level) effect of house price growth on consumption and serves as a control, so that the interaction is not confounded with the triple-interaction term θ_h. The coefficient θ_h is the object of interest in this extension. The triple interaction $s_t \times \widetilde{E}^{ho}_{c,d} \times \widetilde{HP}_{c,t-1}$ measures whether the exposure channel γ_h is amplified or dampened by the state of the housing market. The housing asset share captures the illiquid, valuation-sensitive component of a household portfolio. Higher-exposure households are expected to contract their consumption more.

Figure 6.2a gives the intuitive picture: at short horizons the high-exposure curve (orange) lies

[14]

$$\widetilde{HP}_{c,t-1} = \frac{HP_{c,t-1} - \overline{HP}}{10},$$

where $HP_{c,t-1}$ is the four-quarter log growth of the house price index (in %) lagged one quarter, and $\overline{HP}$ is its pooled mean across country-quarters; the divisor of 10 sets one unit of $\widetilde{HP}$ equal to 10 pp of year-on-year growth.

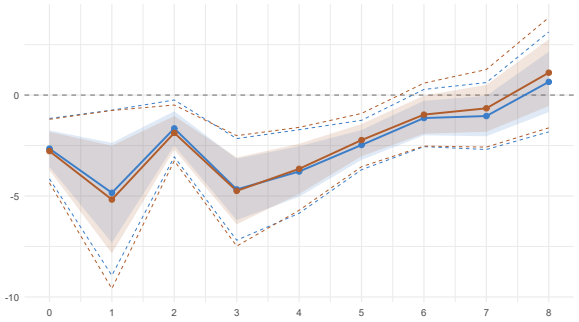
below the low-exposure curve (blue), so housing-rich cells go through a deeper contraction at first, before the two converge as the exposure differential reverses. The gap then closes and switches sign, with the high-exposure curve moving above the low-exposure one at longer horizons as these cells recover. Figure 6.2b shows that the exposure differential is negative and significant at 68% on impact, then turns positive and significant at $h = 5, 7, 8$. This later reversal is consistent with the collateral channel working in the opposite direction: as the tightening fades and expected house prices recover, the borrowing constraint on housing-rich households loosens, and the cells that were most constrained on impact are the ones that can borrow and spend again, so they rebound by more than the average. We do not build on it, however, the reversal is significant only at the 68% level. At the horizon of deepest relative contraction ($h = 1$), moving from the 25th to the 75th percentile of housing exposure deepens the contraction by around 0.33 pp, about 7% of the average response at that horizon. This contraction is consistent with the same portfolio logic as the liquidity LP, now with the opposite sign: following a rise in the interest rate, households with more wealth concentrated in housing cut consumption more.

The additive house price coefficient ρ_h of Table 6.2 is positive and significant at every horizon, rising over the first six quarters. This is intuitive because stronger price growth raises cumulative consumption growth, consistent with a housing wealth effect. However, ρ_h is entered as a control, so that the triple interaction θ_h isolates the interaction rather than confounding it with the direct housing-wealth effect.

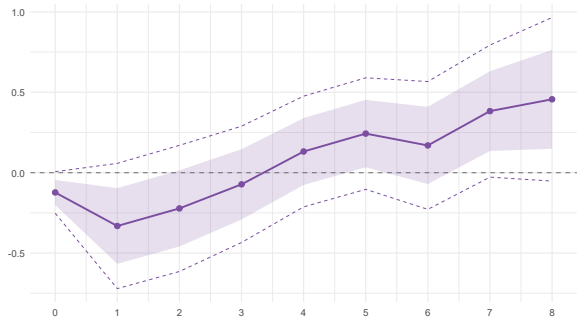
The triple interaction θ_h asks whether the effect of housing exposure on the consumption response depends on house price growth. In Figure 6.2c, θ_h is not significant at any horizon. Therefore we turn to Figure 6.2d, which plots the implied exposure slope at low (P25) and high (P75) house price growth. The two slopes follow a very similar pattern. In both housing-market states, the exposure slope is negative at short horizons, meaning that higher housing exposure is associated with a deeper initial consumption contraction. From the medium horizons onward, the slope turns positive, consistent with the recovery pattern already visible in the differential response. However, the low- and high-house price-growth slopes are close to each other and cross over time, confirming the weak state-dependence already suggested by the insignificant θ_h. The evidence points to an exposure effect that is concentrated at short horizons, rather than one that is systematically stronger when house prices had previously been rising.

These patterns are consistent with a valuation channel operating through housing wealth.

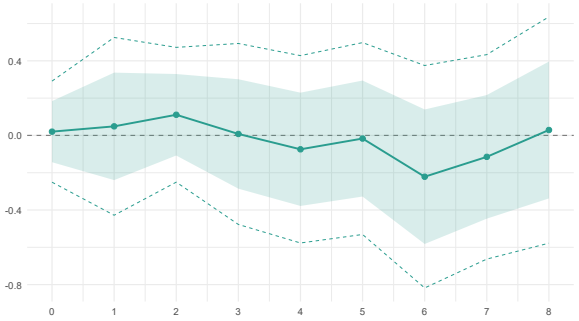
Households whose wealth is concentrated in housing hold an illiquid, long duration, and valuation-sensitive asset. After a tightening, discount rates rise and house prices are depressed, eroding net worth and the collateral against which they borrow. Unable to draw on liquid assets and facing tighter credit, they cut back even more on consumption than the average and recover as the constraint later loosens. This holds regardless of the state of the housing market.



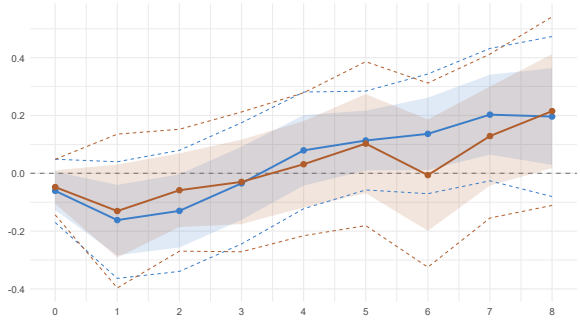
(a) By housing asset-share exposure



(b) Differential response



(c) Triple interaction θ_h



(d) Exposure-channel slope by house price state

Figure 6.2: **Housing asset share exposure and consumption response to a monetary policy shock.**

Notes. In all panels the horizontal axis is the horizon h (quarters). The vertical axis is the cumulative log change in real consumption (pp), except in (c) where it is θ_h. (a): response at low (blue, P25, $\widetilde{E}^{ho} = -0.90$) versus high (orange, P75, $\widetilde{E}^{ho} = 1.33$) housing exposure. (b): differential response $\gamma_h \times (\widetilde{E}^{ho}_{75} - \widetilde{E}^{ho}_{25})$. (c): triple-interaction coefficient θ_h: shock × house share × lagged year-on-year house price growth. (d): exposure slope $\gamma_h + \theta_h \widetilde{HP}_{c,t-1}$ at low (blue, $\widetilde{HP} = -0.27$, P25) versus high (orange, $\widetilde{HP} = 0.37$, P75) house price growth. Shaded band: 68% CI. Dashed lines: 90% CI.

Table 6.2: **Local projection estimates, housing exposure.** Response of consumption to a 25 bp contractionary shock, with house price interaction.

	β_h		γ_h		ρ_h		θ_h	
h	Coef.	S.e.	Coef.	S.e.	Coef.	S.e.	Coef.	S.e.
0	−2.701*	0.928	−0.055†	0.035	1.466*	0.561	0.020	0.165
1	−4.976*	2.566	−0.149†	0.106	2.312*	0.762	0.049	0.290
2	−1.741*	0.842	−0.100	0.107	3.086*	1.003	0.111	0.220
3	−4.703*	1.581	−0.033	0.098	3.732*	0.950	0.008	0.295
4	−3.733*	1.250	0.059	0.094	4.323*	1.018	−0.074	0.305
5	−2.375*	0.761	0.109†	0.094	4.595*	1.067	−0.017	0.313
6	−1.069†	0.889	0.076	0.108	4.692*	1.048	−0.222	0.363
7	−0.885	1.069	0.172†	0.112	4.510*	0.948	−0.115	0.333
8	0.837	1.562	0.204†	0.139	4.278*	0.868	0.029	0.369

Notes. * $|t| \geq 1.645$ (90%); † $|t| \geq 1$ (68%). Driscoll–Kraay SE. The fraction of missing information from multiple imputation is below 0.01 for all coefficients at every horizon.

6.3 Equity Exposure and the Valuation Channel

This specification is the equity asset version of Equation (2): the exposure measure $\widetilde{E}^{e}_{c,d}$ is now the household equity share, and the housing-market terms are replaced by their share price counterparts:

$$y_{c,t+h} = \alpha_c + \beta_h s_t + \gamma_h \left(s_t \times \widetilde{E}^{e}_{c,d} \right) + \rho_h \widetilde{SP}_{c,t-1} + \theta_h \left(s_t \times \widetilde{E}^{e}_{c,d} \times \widetilde{SP}_{c,t-1} \right) + \boldsymbol{\delta}_h' \mathbf{X}_{c,t} + \varepsilon_{c,d,t+h}, \quad (3)$$

Here $\widetilde{SP}_{c,t-1}$ is the quarter-on-quarter log return of the national share price index (OECD 2026), lagged one quarter, centered and scaled as $\widetilde{HP}_{c,t-1}$[15]. As in the housing specification, ρ_h controls for the direct effect of share price growth and θ_h is the object of interest, now measuring whether the equity channel is amplified or dampened by the state of the equity market.

Figure 6.3a reports the response at low and high equity exposure. The high-exposure curve (orange) lies above the low-exposure one (blue), so cells with a larger equity share cut consumption by less after a tightening. The gap is small and, as the differential makes clear, imprecisely estimated. The differential response point estimate is positive at short-to-medium horizons but only marginally distinguishable from zero using the 1-SE band. It hits its largest magnitude at $h = 3$, where moving from the 25th to the 75th percentile of equity exposure decreases the contraction by around 0.20 pp, about 4.4% of the average response at this horizon. The point

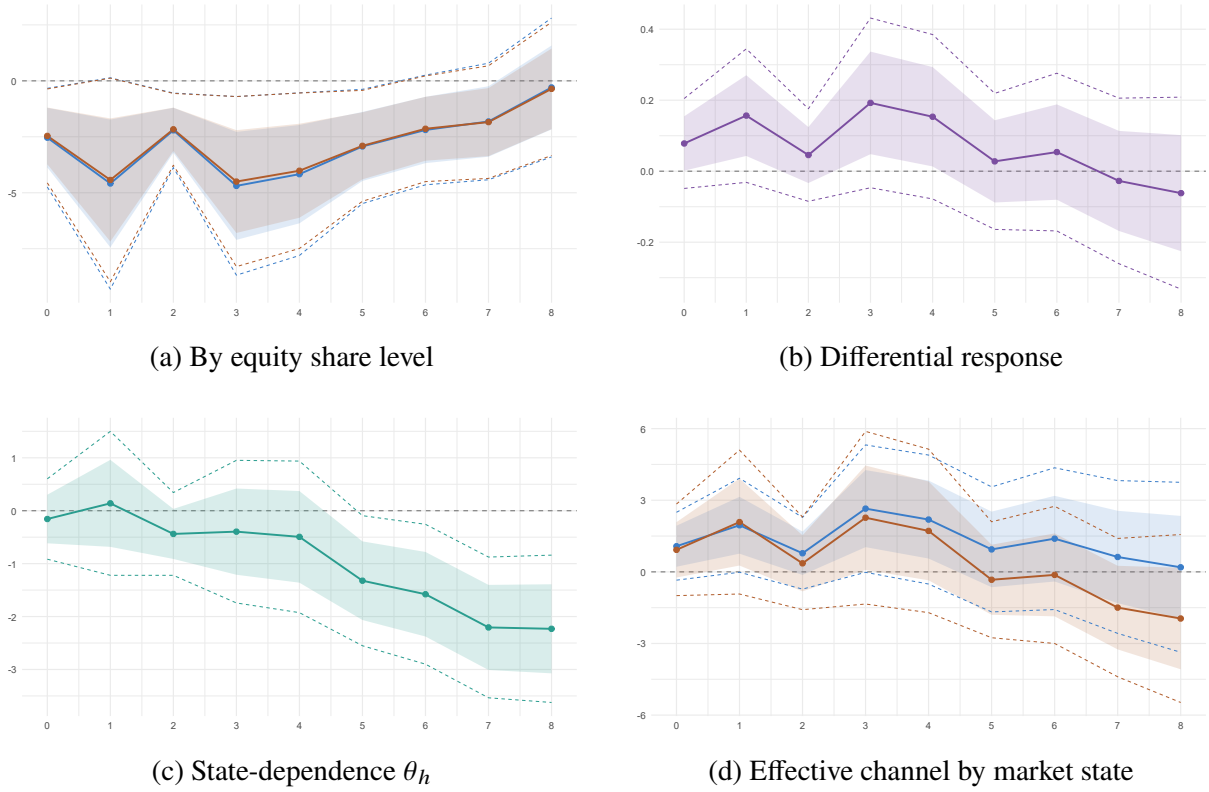
[15]The equity share is scaled by 10 pp, as for the other two exposure measures. The inter-quartile range is 0.8 pp. The coefficient γ_h is therefore reported at a unit well outside the range of the data and is not comparable in magnitude to the liquidity and housing coefficients, which are scaled the same way but span a much wider range. The relevant object for comparison is the differential response.

estimates thus suggest that, following a rise in the interest rate, holders of equity cut their consumption slightly less. However, the unconditional channel is weak and its role here is mainly to set up the state-dependent result below.

θ_h in Figure 6.3c is significant at 90% from $h = 5$ onward. It captures that after a period of rising equity prices, the response of a high-equity cell to a monetary tightening becomes more negative, increasingly so at longer horizons. This points to a slow adjustment of consumption to perceived wealth rather than a reaction on impact. Households hit by a tightening after a strong market quarter cut consumption more, but only with a delay of several quarters.

Figure 6.3d shows two slopes that are close together and both positive at short horizons. From the medium horizons onward they separate, the high-return slope (orange) falls below the low-return slope (blue) and turns negative from around $h = 5$, while the other remains positive. By $h = 7 - 8$ the gap between the two market states reaches roughly 2 pp. When a monetary tightening follows a period of strong share price growth, higher equity exposure is therefore associated with a larger drop in consumption. The additive share price coefficient ρ_h is negative and marginally significant on impact, close to zero and insignificant over the first four quarters, and positive and significant at the 90% level from $h = 5$ onward. We do not over-read it, since it enters only as a control, but the positive estimates at longer horizons are consistent with stronger prior equity price growth being associated with a higher subsequent consumption path.

The result to retain from this section is the negative and significant θ_h at medium and long horizons, which indicates that the equity channel is state-dependent: the effect of equity exposure on consumption is not constant, but varies with the prior stock-market return. Equity exposure cushions the response in ordinary times: at short horizons, and after average or weak prior returns, equity-exposed cells cut consumption by less. The cushion is eroded, and eventually reversed, when the tightening follows a period of strong price growth, since a larger stock of unrealised valuation gains is then revalued downwards. The mechanism is consistent with a discount-rate and revaluation channel operating through portfolio wealth rather than cash flow, which acts slowly and conditionally on the market state entering the shock. Read alongside the previous results, the timing of the three channels lines up with the payoff horizon ordering of the underlying assets: the liquidity channel operates on impact, the housing channel over the first quarters, and the equity channel only from the second year onward.



(a) By equity share level

(b) Differential response

(c) State-dependence θ_h

(d) Effective channel by market state

Figure 6.3: **Equity share exposure and the state-dependent equity channel.**

Notes. In all panels the horizontal axis is the horizon h (quarters). The vertical axis is the cumulative log change in real consumption (pp), except in (c) where it is θ_h. Shaded band: 68% CI. Dashed lines: 90% CI. (a): response at low (blue, P25, $\widetilde{E}^e = -0.06$) versus high (orange, P75, $\widetilde{E}^e = 0.02$) equity share exposure. (b): differential response $\gamma_h \times (\widetilde{E}^e_{75} - \widetilde{E}^e_{25})$. (c): triple-interaction coefficient θ_h: equity share × prior-quarter share price return. (d): effective equity channel $\gamma_h + \theta_h \widetilde{SP}_{c,t-1}$ at low (blue, P25, $\widetilde{SP} = -0.44$) versus high (orange, P75, $\widetilde{SP} = 0.52$) prior-quarter share price return.

Table 6.3: **Local projection estimates, equity exposure.** Response of consumption to a 25 bp contractionary shock, with share price interaction.

	β_h		γ_h		ρ_h		θ_h	
h	Coef.	S.e.	Coef.	S.e.	Coef.	S.e.	Coef.	S.e.
0	−2.477*	1.289	1.004†	0.990	−0.185†	0.174	−0.157	0.461
1	−4.463†	2.789	2.014†	1.470	−0.262	0.286	0.139	0.827
2	−2.176*	0.983	0.585	1.019	0.085	0.139	−0.439	0.475
3	−4.543*	2.335	2.474†	1.868	−0.037	0.251	−0.395	0.819
4	−4.050*	2.132	1.972†	1.810	0.107	0.249	−0.494	0.871
5	−2.909*	1.515	0.356	1.498	0.489*	0.289	−1.322*	0.747
6	−2.153†	1.446	0.693	1.736	0.443*	0.191	−1.578*	0.803
7	−1.838†	1.540	−0.352	1.822	0.604*	0.269	−2.205*	0.808
8	−0.345	1.825	−0.795	2.114	0.786*	0.304	−2.232*	0.846

Notes. * $|t| \geq 1.645$ (90%); † $|t| \geq 1$ (68%). Driscoll–Kraay SE. The fraction of missing information from multiple imputation is below 0.01 for all coefficients at every horizon.

7 A FOUR AGENT NEW KEYNESIAN MODEL

7.1 Model

Households. Our model builds from the TANK model of Galí et al. (2007) and Bilbiie (2008). We consider an economy with four types of infinitely-lived households indexed by $i \in \{1, 2, 3, 4\}$ and present in constant population shares ω_i with $\sum_i \omega_i = 1$. All households supply labour and consume a single final good, they differ in the asset they hold and in labour productivity z_i. They share the same preference specification over consumption and hours, the labour-disutility weight $\psi_{N,i}$ is agent-specific, and housing enters utility only for agent 3, with weight ψ_H:

$$E_t \sum_{s=0}^{\infty} \beta_i^s \left[\frac{C_{i,t+s}^{1-\sigma}}{1-\sigma} - \psi_{N,i} \frac{N_{i,t+s}^{1+\varphi}}{1+\varphi} + \mathbf{1}\{i = 3\} \psi_H \log H_{3,t+s} \right], \tag{1}$$

where σ is the inverse elasticity of intertemporal substitution (EIS) and φ the inverse Frisch elasticity of labour supply. $\psi_{N,i}$ weights the disutility of labour and is agent-specific, calibrated so that all types supply the same hours in steady state (Section 7.2). The discount factor β is common to all agents except for the impatient borrower agent 3 ($\beta_3 < \beta$). Agent 1 is hand-to-mouth (Bilbiie 2008), it holds no assets and consumes all of its labour income each period. Agent 2 saves through a one-period bond subject to a quadratic portfolio adjustment cost. Agent 3 owns housing and borrows against it, it is impatient ($\beta_3 < \beta$) so that the collateral constraint binds (Iacoviello 2005). Agent 4 owns the firms and prices the resulting equity claim with its own stochastic discount factor. The bond is issued by agent 3 and held entirely by agent 2, while housing and equity stocks are in fixed supply. The four types thus embody four distinct transmission channels: the current-income channel of agent 1, the intertemporal-saving channel of agent 2, the collateral and housing-wealth channel of agent 3, and the valuation channel of agent 4.

Consumption and saving. The four types differ in the assets they hold, in whether and how those assets let them shift consumption across time, and, for agent 3, in patience. **Agent 1** holds no assets and consumes its labour income each period, without a means of saving it cannot substitute intertemporally after a movement in the wage.

$$C_{1,t} = W_t z_1 N_{1,t}, \tag{2}$$

Agent 2 saves through a one period nominal bond, $b_{2,t} = B_{2,t}/P_t$ denotes the real value of holdings, and is subject to an adjustment cost $\frac{\eta}{2}\left(b_{2,t} - \bar{b}\right)^2$ when it differs from the steady state position $\bar{b}$. We adopt the quadratic form of Schmitt-Grohé and Uribe (2003) because it induces a determinate steady-state bond position and it represents a reduced-form friction, so that agent 2 can smooth shocks only partially. Its budget constraint (3) and Euler (4) read:

$$C_{2,t} + b_{2,t} + \frac{\eta}{2}\left(b_{2,t} - \bar{b}\right)^2 = \frac{1 + i_{t-1}}{1 + \pi_t}\, b_{2,t-1} + W_t z_2 N_{2,t}, \tag{3}$$

$$C_{2,t}^{-\sigma}\left[\, 1 + \eta\left(b_{2,t} - \bar{b}\right)\right] = \beta\, E_t\left[C_{2,t+1}^{-\sigma}\, \frac{1 + i_t}{1 + \pi_{t+1}}\right]. \tag{4}$$

Agent 3 borrows against the value of its housing stock, limited by a collateral constraint (5), and subject to a budget constraint (6):

$$d_{3,t}\,(1 + i_t) = m_H\, E_t\left[q_{t+1}^h\,(1 + \pi_{t+1})\right] H_{3,t}, \tag{5}$$

$$C_{3,t} + \frac{1 + i_{t-1}}{1 + \pi_t}\, d_{3,t-1} + q_t^h\left[H_{3,t} - (1 - \delta_H)H_{3,t-1}\right] = W_t z_3 N_{3,t} + d_{3,t}, \tag{6}$$

where $d_{3,t}$ is the real value of its debt and δ_H the depreciation rate of housing. The collateral constraint limits how much debt can be taken at a fraction m_H of the expected future value of housing it holds. An expected fall in prices tightens the constraint. Since $\beta_3 < \beta$, agent 3 always borrows up to the limit and the constraint binds, the multiplier μ_t therefore appears in both first-order conditions, on housing (7) and debt (8):

$$C_{3,t}^{-\sigma} q_t^h = \beta_3 E_t\left[C_{3,t+1}^{-\sigma} q_{t+1}^h (1 - \delta_H)\right] + \frac{\psi_H}{H_{3,t}} + \mu_t\, m_H\, E_t\left[q_{t+1}^h (1 + \pi_{t+1})\right]. \tag{7}$$

$$C_{3,t}^{-\sigma} = \beta_3\, E_t\left[C_{3,t+1}^{-\sigma} \frac{1 + i_t}{1 + \pi_{t+1}}\right] + \mu_t\,(1 + i_t). \tag{8}$$

Equation (7) shows the cost of an additional unit of housing against its value next period, the housing utility it provides and the additional borrowing it allows valued at μ_t. A fall in expected housing price decreases borrowing capacity, so agent 3 is poorer. The shadow value of relaxing the borrowing limit by one unit is $\mu_t(1 + i_t)$. Agent 3 is affected by the policy rate through the intertemporal price and also through the value of the collateral constraint itself.

Agent 4 holds the equity claim on the firms, whose budget constraint (9) and Euler equation (10)

are:

$$C_{4,t} + q_t^e S_{t+1} = \left(q_t^e + d_t\right) S_t + W_t z_4 N_{4,t}. \tag{9}$$

$$q_t^e = \beta\, E_t\left[\left(\frac{C_{4,t+1}}{C_{4,t}}\right)^{-\sigma}\left(q_{t+1}^e + d_{t+1}\right)\right]. \tag{10}$$

where S_t denotes the shares carried into period t. Equation (10) prices the equity claim through the expected payoff next period, discounted by agent 4's valuation of future consumption. The same discount factor enters the firms' pricing decision since agent 4 owns them. Because the equity stock is fixed and held entirely by agent 4, the valuation is not realised through a sale. A rise in the discount rate lowers the price q_t^e even when d_t is unchanged ($d_t = Y_t - W_t N_t - \frac{\theta}{2}\pi_t^2 Y_t$). Monetary policy affects this type primarily by repricing its portfolio.

Labour supply. All four types face the same competitive wage per efficiency unit W_t. Agent i supplies $z_i N_{i,t}$ and earns $W_t z_i N_{i,t}$, with the productivities per agent normalised so that $\sum_i \omega_i z_i = 1$. Firms hire the aggregate input $N_t = \sum_i \omega_i z_i N_{i,t}$ and do not distinguish across types. The intratemporal first-order condition equates the marginal rate of substitution between consumption and leisure to the wage:

$$\psi_{N,i}\, N_{i,t}^{\varphi}\, C_{i,t}^{\sigma} = W_t z_i, \qquad i = 1, \ldots, 4. \tag{11}$$

Types differ in their labour supply through their productivity and through the consumption path implied by their own portfolio.

Firms. In this economy there is a perfectly competitive final-good producer using CES with elasticity of substitution ϵ, that aggregates a continuum of differentiated varieties $j \in [0, 1]$ into the single good that households consume (Galí 2015). Each variety is produced by a monopolistically competitive firm using only labour, $Y_{j,t} = A_t N_{j,t}$, so real marginal cost is common across firms, $mc_t = W_t / A_t$. Intermediate firms' pricing decisions are subject to quadratic Rotemberg adjustment cost $\frac{\theta}{2}\pi_t^2 Y_t$ (Rotemberg 1982). To a first-order approximation this formulation yields the same Phillips curve as Calvo pricing and is simple to handle (Galí 2015). Since all impulse responses below are first-order, the equivalence is exact for our purposes. Real profit is output net of the wage paid to workers and the price-adjustment cost, redistributed to shareholders (agent 4) as the dividend d_t. The resulting symmetric equilibrium price-setting

condition is the New Keynesian Phillips curve (12):

$$\theta\,\pi_t(1+\pi_t) = (1-\epsilon) + \epsilon\, mc_t + \theta\,\beta\, E_t\left[\left(\frac{C_{4,t+1}}{C_{4,t}}\right)^{-\sigma} \pi_{t+1}(1+\pi_{t+1})\,\frac{Y_{t+1}}{Y_t}\right]. \tag{12}$$

Current inflation depends on real marginal cost and on expected future inflation, with the slope governed by θ: a higher adjustment cost makes prices more rigid and flattens the curve. The shock is absorbed by output rather than prices, making monetary policy more powerful in real terms.

Monetary policy. The central bank sets the nominal rate according to the Taylor rule (Taylor 1993):

$$1 + i_t = \frac{1}{\beta}\,(1+\pi_t)^{\phi_\pi}\left(\frac{Y_t}{\bar{Y}}\right)^{\phi_y} \exp(v_t), \tag{13}$$

with $\bar{Y}$ the steady state value of Y, $\phi_\pi > 1$ the response to inflation, $\phi_y > 0$ the response to output, and v_t the monetary policy shock. Productivity and policy shock follow AR(1) processes:

$$\log A_t = \rho_a \log A_{t-1} + \varepsilon_t^a, \qquad v_t = \rho_v v_{t-1} + \varepsilon_t^v, \tag{14}$$

with ε_t^a and ε_t^v i.i.d. and mutually independent. A positive ε_t^v constitutes an unanticipated monetary tightening, the shock traced in the impulse responses of Section 7.3.

Market clearing. Labour supply aggregates across types in efficiency units, $N_t = \sum_i \omega_i z_i N_{i,t}$, and output follows from the linear technology, $Y_t = A_t N_t$. The bond market clears internally, $\omega_2 b_{2,t} = \omega_3 d_{3,t}$, so that the debt issued by agent 3 is held entirely by agent 2 and net supply is zero. The equity and housing stocks are in fixed supply, $\omega_4 S_t = 1$ and $\omega_3 H_{3,t} = \bar{H}$. Goods market clearing is:

$$Y_t = \sum_{i=1}^{4} \omega_i C_{i,t} + \frac{\theta}{2}\pi_t^2 Y_t + \omega_2 \frac{\eta}{2}\left(b_{2,t} - \bar{b}\right)^2, \tag{15}$$

By Walras's law, (15) is redundant and is dropped from the solved system. The model is written in levels and solved by first-order approximation around its deterministic steady state, which is reported in Section 7.2[16].

[16] Interest rates and inflation are instead reported as deviations of the quarterly rate, annualised and in basis points, since a percentage deviation is uninformative for a rate and undefined at $\bar{\pi} = 0$.

7.2 Calibration

The model is calibrated at a quarterly frequency and to Germany, one of the ten countries studied in Section 5. Parameters fall into three groups: borrowed from the standard New-Keynesian and housing literature, measured directly from household-level data (European Central Bank 2022), and calibrated internally to match steady-state targets.

7.2.1 The Wealth Partition

The four agents are assigned weights $(\omega_1, \ldots, \omega_4) = (0.30, 0.20, 0.40, 0.10)$ corresponding to bands of the net-wealth distribution, for example agent 4 corresponds to the top decile. The partition is imposed, each band holding the portfolio of one agent: liquid claims in lower-middle, housing and mortgage debt in the upper-middle and equity at the top. German households sort more closely to this partition than those of other euro area countries. Table 7.1 reports the composition of gross assets within each band and supports our choice.

Table 7.1: **Portfolio composition by net-wealth band.** Germany, HFCS Wave 4.

	Agent 1	Agent 2	Agent 3	Agent 4
Share of gross assets (%)				
Main residence	22.0	25.6	60.2	33.0
Deposits	25.5	32.1	12.2	6.3
Listed shares	0.6	2.0	1.9	3.8
Mortgage debt / gross assets	54.6	17.5	9.8	4.3
Participation (%)				
Holds equity	2.9	10.7	19.8	45.1
Holds a mortgage	2.0	9.8	30.6	32.2

Deposits are highest for agent 2 and the main residence share for agent 3, while listed shares are thin everywhere as a portfolio share, but participation in equity is concentrated at the top, reaching 45% for agent 4. Germany is preferred to France because this separation is sharper, wider French homeownership spreads housing across the bands and blurs the distinction between saver and borrower. The weak point in Germany is agent 1, which holds a mortgaged residence rather than nothing. However, the band's net position is near zero, it holds 0.2% (Table 7.2) of aggregate net wealth against 30% of the population. This is consistent with its portfolio, in which mortgage debt equals 54.6% of gross assets, so that much of what the band owns is debt-financed

and its liquid wealth is too thin to buffer a shock (Kaplan, Violante, and Weidner 2014).

7.2.2 Preferences and Technology

Following the standard New-Keynesian calibration (Galí 2015), the discount factor of the patient savers is $\beta = 0.99$, and preferences are logarithmic in consumption ($\sigma = 1$). The inverse Frisch elasticity is set to $\varphi = 0.01$, so that labour supply is nearly perfectly elastic and hours adjust at almost no wage premium: the real wage then tracks consumption alone, which dampens the response of real marginal cost $mc_t = W_t$. This value is a reduced form for wage rigidity, in the spirit of the near-linear labour disutility of Iacoviello (2005)[17]. The elasticity of substitution across varieties is $\epsilon = 6$, giving a 20% markup. The borrower (agent 3) is impatient, $\beta_3 = 0.97 < \beta$, so that it always prefers to borrow up to its collateral limit rather than save, keeping the constraint binding. Housing does not depreciate and so behaves like land, its value derives from the service flow and its collateral role, not from production (Iacoviello 2005). The Rotemberg cost is set to $\theta = 58.25$: around a zero-inflation steady state, Rotemberg and Calvo pricing imply the same first-order Phillips curve (Roberts 1995), so θ can be calculated from a Calvo probability α:

$$\theta = \frac{(\epsilon - 1)\,\alpha}{(1-\alpha)(1-\alpha\beta)}, \tag{16}$$

following Galí (2015), $\alpha = 0.75$ implies an average price duration of four quarters, broadly consistent with euro area micro evidence of infrequent price adjustment (Dhyne et al. 2006).

7.2.3 Housing and the Collateral Constraint

The housing weight ψ_H is calibrated, following Iacoviello (2005), so that the model reproduces the ratio of housing wealth to income observed for Germany. This ratio is 3.559 in the HFCS (European Central Bank 2022) and is expressed relative to annual income, since the model is quarterly, annual income is $4\bar{Y}$, so the steady-state target is $q^h\bar{H} = 3.56\,(4\bar{Y})$, and matching it gives $\psi_H = 1.01$. The loan-to-value ratio m_H is measured as aggregate German mortgage debt over aggregate housing wealth. The bond adjustment cost is set to a small value, $\eta = 0.0025$, rather than to a target. It mainly secures a determinate steady-state bond position and gives

[17] At the conventional value $\varphi = 1$ the wage falls by roughly twice the fall in output, so marginal cost overshoots, the resulting deflation triggers a policy response that lowers the real rate on impact, reversing the sign of the transmission. At $\varphi = 0.01$ the wage falls by about as much as output, dampening the deflation and restoring the sign. Appendix V reports the sensitivity.

agent 2 a mild friction so that it can smooth but slightly less than a frictionless saver.

7.2.4 Labour Supply and Policy

Steady-state hours are normalised to $N_i = 1/3$ for every agent, the standard value in the business cycle literature (Cooley and Prescott 1995), corresponding to roughly one third of discretionary time allocated to market work. Because agents differ in steady-state consumption, this normalisation is achieved through an agent-specific disutility weight $\psi_{N,i} = W z_i/(N_i^{\varphi} C_i^{\sigma})$. Labour income then differs through productivity $z_i = \alpha_i/\omega_i$ where α_i is the share of aggregate gross labour income within each band i in the German HFCS (European Central Bank 2022): 15.1, 17.4, 48.5 and 19.0% for agents 1 to 4 respectively. Equalising hours therefore leaves productivity as the only source of labour-income heterogeneity, so that differences in the response to shocks come from the portfolio and the composition of income alone.

The central bank follows a Taylor rule with $\phi_\pi = 1.50$ and an output response $\phi_y = 0.125$, the quarterly equivalent of the value in Taylor (1993). The productivity and monetary shocks follow AR(1) processes with persistence $\rho_a = 0.90$ and $\rho_v = 0.50$, and innovation standard deviations $\sigma_a = 0.01$ and $\sigma_v = 0.0025$. The model is solved to first order, the size of the monetary innovation only scales the impulse responses, which are normalised to 25 annualised bp.

7.2.5 Steady State and the Wealth Distribution

Table 7.2 compares the resulting net-wealth shares with the data. The concentration at the top and in the middle 40% is reproduced closely without being targeted. The model shares are steady state asset holdings, not fitted moments. The one weakness is a mild overstatement of agent 2 whose position is tied to the borrower's leverage and cannot be set independently.

Table 7.2: **Net-wealth distribution.** Model versus data, Germany, HFCS.

Agent	Model	Data (DE)
Agent 1 (bottom 30%)	0.0%	0.2%
Agent 2 (30–50)	5.6%	3.1%
Agent 3 (50–90)	40.7%	41.0%
Agent 4 (top 10%)	53.7%	55.7%

Table 7.3: **Calibrated parameters.** Quarterly frequency.

Parameter	Description	Value	Source / target
Panel A. Preferences and technology			
β	Discount factor, savers	0.99	Standard, quarterly
β_3	Discount factor, borrower	0.97	Iacoviello (2005)
σ	Inverse EIS (log utility)	1.00	Log utility
φ	Inverse Frisch elasticity	0.01	Iacoviello (2005)
ϵ	Elasticity of substitution	6.00	Markup = 20%
θ	Rotemberg price adjustment	58.25	Calvo $\alpha = 0.75$; eq. (16)
δ_H	Housing depreciation	0.00	Housing as land (Iacoviello 2005)
η	quadratic bond adjustment cost	0.0025	Untargeted; small
Panel B. Population shares (net-wealth partition, DE)			
ω_1	Share, agent 1 (bottom 30%)	0.30	HFCS partition
ω_2	Share, agent 2 (30–50)	0.20	HFCS partition
ω_3	Share, agent 3 (50–90)	0.40	HFCS partition
ω_4	Share, agent 4 (top 10%)	0.10	HFCS partition
Panel C. Labour disutility and productivity			
$\psi_{N,1}$	Disutility weight, agent 1	3.0331	Target $N_1 = 1/3$
$\psi_{N,2}$	Disutility weight, agent 2	2.7065	Target $N_2 = 1/3$
$\psi_{N,3}$	Disutility weight, agent 3	3.1703	Target $N_3 = 1/3$
$\psi_{N,4}$	Disutility weight, agent 4	1.4764	Target $N_4 = 1/3$
z_1	Labour productivity, agent 1	0.5039	HFCS labour-income share
z_2	Labour productivity, agent 2	0.8697	HFCS labour-income share
z_3	Labour productivity, agent 3	1.2131	HFCS labour-income share
z_4	Labour productivity, agent 4	1.8967	HFCS labour-income share
Panel D. Housing and collateral			
ψ_H	Housing preference weight	1.01	Target: housing/income = 3.56
m_H	Loan-to-value ratio	0.1229	HFCS (aggregated DE households)
$\bar{H}$	Housing stock (normalisation)	1.00	Normalisation
Panel E. Policy and shocks			
ϕ_π	Taylor: inflation	1.50	Taylor (1993)
ϕ_y	Taylor: output gap	0.125	Taylor (1993)
ρ_a	Persistence, productivity	0.90	Standard
ρ_v	Persistence, monetary	0.50	Standard
σ_a	Std. dev., productivity shock	0.01	Normalisation
σ_v	Std. dev., monetary shock	0.0025	Scales IRFs

7.3 Results

This section describes the response of a calibrated economy to an unanticipated monetary tightening and asks whether the four types reproduce the channels identified empirically in Section 6. Responses are normalised so that the nominal policy rate rises by 25 annualised bp on impact, the same size as the local-projection experiment, so that the two sets of responses are directly comparable.[18]

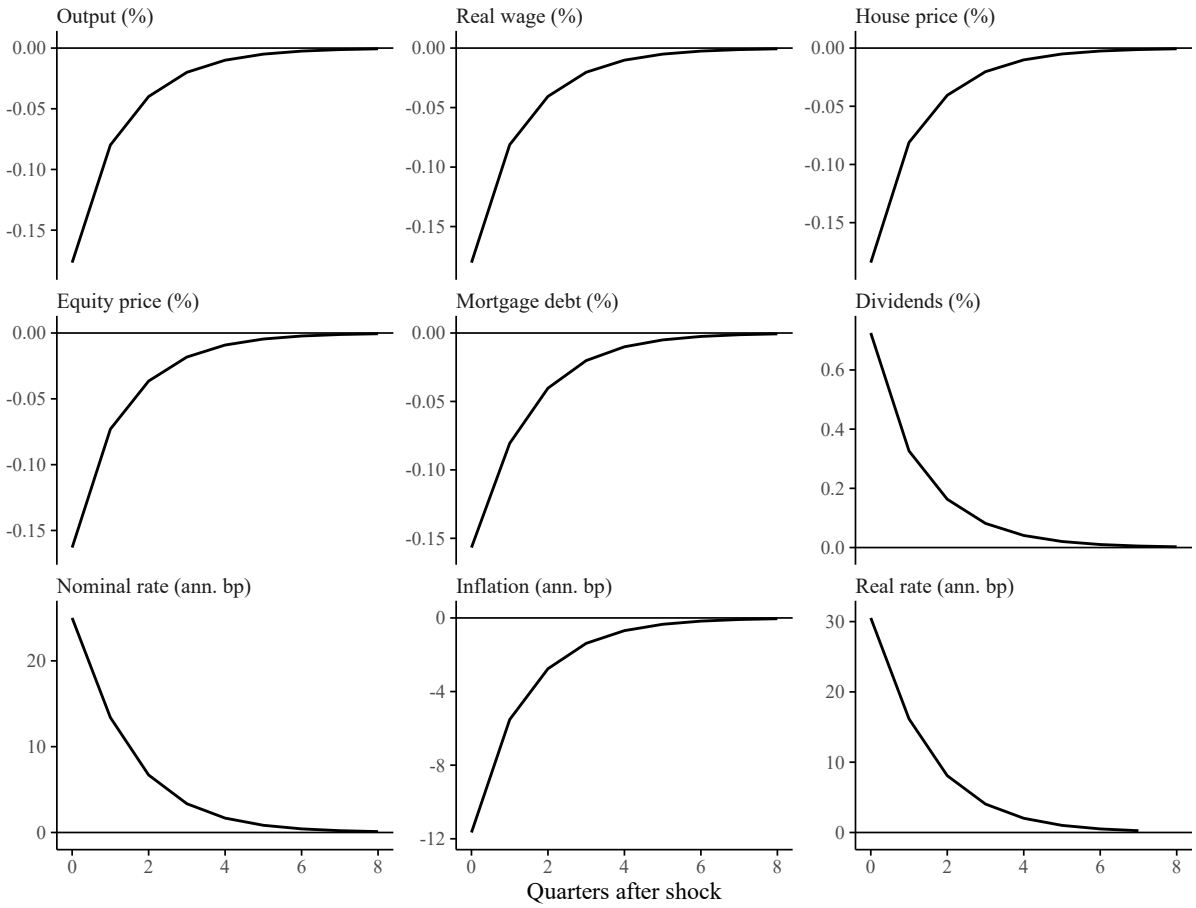


Figure 7.1: **Aggregate impulse responses.** Responses to a contractionary monetary policy shock.

Notes. Responses to a contractionary monetary policy of 25 annualised bp on impact. Quantities and prices are percentage deviations from the deterministic steady state; interest rates and inflation are in annualised bp. The real rate is $r_t = i_t - \mathbb{E}_t \pi_{t+1}$. Hours and marginal cost are omitted since A_t fixed, $Y_t = A_t N_t$ and $mc_t = W_t$.

The aggregate response. Figure 7.1 reports the impulse response to the tightening. The nominal policy rate rises by 25 bp on impact while expected inflation falls, so the expected real rate rises by more than the nominal rate, to 30.5 bp, impact inflation itself falls by 11.7 bp. With sticky prices, the central bank moves the intertemporal price by more than the policy

[18] Dynare reports the response to a one-standard-deviation innovation, $\varepsilon^v = \sigma_v = 0.0025$, which raises the policy rate by 48.9 annualised bp; all responses are scaled by $25/49$.

rate, households face this real rate when they make their choices. They consume less, demand for goods falls, and firms produce less rather than cut prices. Output and hours fall by 0.18 %. Labour supply is close to perfectly elastic, so each type's supply curve is nearly flat, consumption falls and the wage falls with it. Firms save both on the hours they hire and on the price of each hour, and the wage bill contracts by roughly twice as much as output. Every variable from $h = 1$ onward inherits the persistence of the shock, decaying at close to $\rho_v = 0.50$ per quarter. With housing and equity in fixed supply, a small bond friction and no capital, no stock builds up to keep the economy away from steady state once the shock has faded, so the endogenous propagation is weak. The model therefore describes how a tightening is allocated across portfolios, and less about how long it takes to work through. Furthermore, the aggregate fall is not distributed in proportion to population. Agent 4, a tenth of the households, accounts for 31% of the decline in aggregate consumption, because it consumes almost eight times as much as Agent 1 in steady state, while Agent 1, three tenths of the population, accounts for 13% (Table 7.5). The type that contracts most is therefore not the type that drives the aggregate.

Cash flow and liquidity. Agent 1 holds no asset, it therefore has no Euler equation and doesn't face intertemporal substitution. Monetary policy reaches it in general equilibrium through the indirect labour-income channel of Kaplan, Moll, et al. (2018). Higher real rates depress demand, firms produce less, the real wage falls, and agent 1, whose income is the wage and whose marginal propensity to consume is one, fully absorbs the fall. Substituting (2) into (11) under log utility gives $\psi_{N,1} N_{1,t}^{1+\varphi} = 1$: the wage cancels, hours are constant, and consumption is proportional to the real wage at every horizon. For this type, transmission is entirely indirect. Agent 2 holds a one-period nominal bond claim and is the least affected of the four (Figure 7.2a). Table 7.4 shows why: it stands on the receiving end of two transfers from the borrower. The unexpected deflation raises the real value of the bond it holds, since the nominal payoff $(1 + i_{t-1}) b_{2,t-1}$ was fixed before the shock, so with a lower price level than expected, agent 2 buys more goods with the same amount. This is Auclert (2019)'s Fisher channel, worth 0.31% of agent 2's consumption. In addition, the borrower's deleveraging arrives at agent 2 as repayment with $\omega_2 b_{2,t} = \omega_3 d_{3,t}$. Agent 2 has no other store of value, the only asset it can hold is fixed by agent 3's collateral constraint. Together the two transfers offset most of the wage loss. Both are exactly mirrored in agent 3's budget and, weighted by population, each cancels to zero, so the two channels redistribute between the types without moving aggregate consumption. The model therefore reproduces the sign of the empirical liquidity result, but agent 2, the relatively liquid saver, is

cushioned not because it draws down its assets, but because its cushion depends on inflation surprise and on agent 3. A regression on portfolio composition cannot separate smoothing from redistribution, since the two produce the same sign. The model can, and it shows that redistribution alone is enough to generate it.

Housing and collateral. Agent 3 contracts the most, and the fall is driven by the asset side of its balance sheet. The house price falls by 0.18%, mortgage debt by 0.16% and the multiplier on the borrowing constraint rises by 1.6%. On one side the collateral is worth less because the tightening depresses the expected house price, and the constraint is written on that price so the amount agent 3 may borrow falls with it. On the other, the outstanding debt is serviced at a higher real rate, so both leave less to consume. Table 7.4 shows the reduced borrowing (-0.702) and the Fisher revaluation of the debt (-0.132) driving the contraction, partly offset by a rise in labour income ($+0.645$) as Agent 3, now poorer, supplies more hours. Of the two sub-mechanisms we presented in Section 1, only the collateral effect operates, a wealth effect would require Agent 3 to trade housing and it does not, since the stock is fixed by market clearing and $\delta_H = 0$. Monetary policy reaches its consumption only through the borrowing limit, so the housing share of Section 6.2, which selects households on which both mechanisms operate, captures more than the model does. However, the direction and the timing match the empirical housing result. The coefficient γ_h is negative on impact and at $h = 1$ before losing significance, and the model produces the same short-horizon pattern through the collateral channel alone. This happens because the borrowing limit tightens once when the shock lands, and the household adjusts once rather than gradually.

Equity and valuation. Equity is a claim to all future dividends, held entirely by agent 4. The tightening depresses the real wage, so agent 4 is poorer today than it expects to be in a few quarters, so it discounts future payments more heavily and the equity price falls by 0.16%. Firms are meanwhile more profitable, the tightening cuts revenue by 0.18% but the wage bill by 0.36%, since employment also falls, and with sticky prices the markup widens, so the aggregate dividend rises by 0.73%. Resources are reallocated from the wage bill to the profit share, and agent 4 holds the claim to it. The equity price falls despite the higher dividend because the rise in the discount rate outweighs the flow gain; the price, however, never enters Agent 4's budget, since the equity stock is fixed and never traded, so its consumption responds only through the dividend flow and labour income. The dividend funds 51% of Agent 4's consumption in steady state and

contributes +0.37 pp on impact. The fall in labour income, which funds the remainder, is of 0.54 pp, leading to a decline in consumption of 0.17%. The loss on wages then exceeds the gain on dividends. Agent 4 is also the only type still distinguishable after impact. By the following quarter, the one-off deflation windfall and repayment have faded and Agents 1–3 have converged, whereas the dividend is a flow that persists for as long as the markup stays wide (Figure 7.2c). The equity differential of Section 6.3 is positive across most horizons, so cells holding more equity contract less, as Agent 4 does here. The state-dependence that is the section's main equity result has no counterpart in a first-order solution.

Model and data. The borrower contracts most, the bond saver least and the hand-to-mouth agent and the shareholder lie in between, the ordering is identical on impact and cumulated over eight quarters (Table 7.5). The three signs the model delivers are the same as the projections deliver. The relatively liquid saver contracts by less than average, as the liquidity differential γ_h of the liquidity index LP is positive; the housing borrower contracts by more, as γ_h is negative on impact and at $h = 1$ in the housing LP; and the equity holder contracts by less, as the unconditional differential of the equity LP is positive across most horizons. None of the three was targeted in the calibration, which fixes population shares, labour-income shares, the housing-to-income ratio and the loan-to-value ratio, and no consumption response. The calibration fixes portfolios, not responses, so the ordering is an output of the mechanisms rather than an imposed feature, though the choice of the four types is itself informed by the empirical section.

Table 7.4: **Impact consumption response.** Decomposed by budget-constraint term.

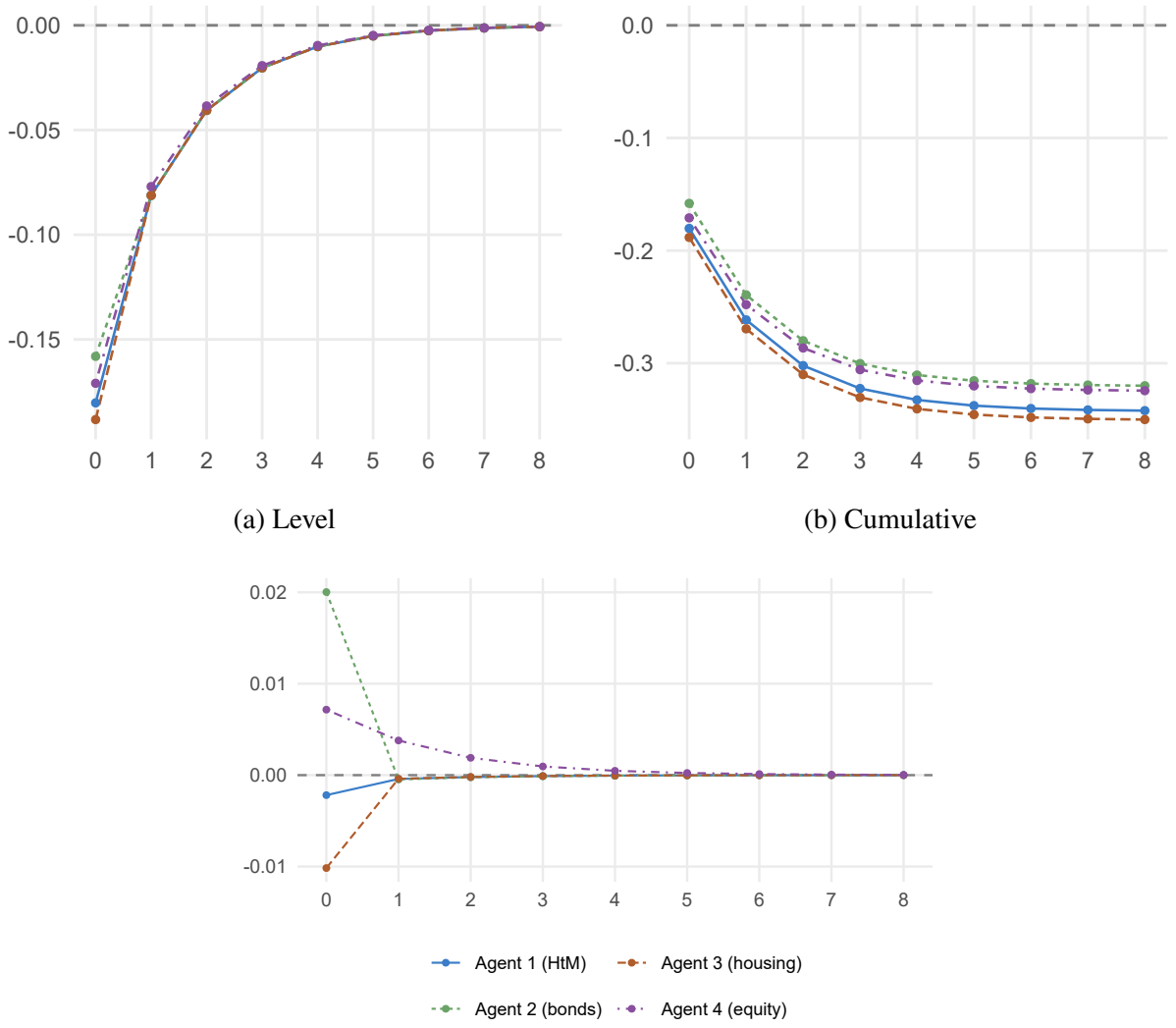
Agent	Labour income	Fisher revaluation	Asset position	Dividends	Total
1. Hand-to-mouth	−0.180	—	—	—	−0.180
2. Illiquid bond holder	−2.143	+0.314	+1.671	—	−0.158
3. Collateral-constrained	+0.645	−0.132	−0.702	—	−0.188
4. Equity holder	−0.543	—	—	+0.372	−0.171

Notes. Each row rebuilds agent i's impact consumption response from its own budget constraint, (2)–(9), as a percentage of that agent's steady-state consumption; the totals match Table 7.5. Labour income is $W_t z_i N_{i,t}$ and differs from the real wage response because hours differ across types. Fisher revaluation is the effect of the inflation surprise on the real value of the nominal claim ($i = 2$) or the nominal debt ($i = 3$). Asset position is the change in bond holdings ($i = 2$) or outstanding debt ($i = 3$). Dividends is $d_t S_t$.

Table 7.5: **Consumption response by agent.**

Agent	ω_i	Impact ($h = 0$) (%)	Cumulative $h = 0{-}8$ (%)	Share of aggregate
1. Hand-to-mouth	0.30	−0.180	−0.342	12.9%
2. Illiquid bond holder	0.20	−0.158	−0.320	14.5%
3. Collateral-constrained	0.40	−0.188	−0.350	41.2%
4. Equity holder	0.10	−0.171	−0.324	31.4%
Aggregate output Y		−0.177	−0.336	100.0%

Notes. Responses to a contractionary monetary policy shock of 25 annualised bp on impact. *Impact* and *Cumulative* are percentage deviations of agent i's consumption from its own steady state, on impact and summed from $h = 0$ through $h = 8$. *Share of aggregate* is agent i's contribution to the fall in aggregate consumption, $\omega_i \bar{C}_i \hat{c}_i$ normalised to sum to one, weighting the percentage response by population share and steady-state consumption. By Walras's law the total equals the output response.



(a) Level

(b) Cumulative

(c) Differential

Figure 7.2: **Consumption responses by agent.** To a contractionary monetary policy shock.

Notes. Responses to a contractionary monetary policy shock of 25 annualised bp on impact. The horizontal axis is the horizon h in all panels. **(a)** percentage deviations from each agent's own steady state, spanning 0.030 pp on a mean of 0.178. **(b)** cumulates panel (a) over h. **(c)** each agent's deviation minus the population-weighted average, which sums to zero across agents at every horizon by construction; a positive value means the agent contracts by less than the average household.

8 CONCLUDING REMARKS

This paper asks whether portfolio composition produces differential, ordered consumption responses to a common euro area monetary policy shock, and whether a four agent New Keynesian model can reproduce that pattern. The answer to both is yes, though the model reproduces the ranking of responses rather than their timing. Households facing the same ECB tightening follow different consumption paths depending on what their wealth is held in, and the differences line up along the payoff horizon of the underlying assets, which is the order in which each channel transmits.

Three main results emerge. Liquidity cushions the response. Households with more liquid portfolios cut consumption by less after a tightening, by about 0.45 pp at the one-quarter horizon, and the effect is transitory and fades within six quarters. Housing exposure shows the opposite pattern. Households whose wealth is concentrated in real estate deepen the contraction, cutting by around 0.33 pp more one quarter out, and this differential does not depend on the state of the housing market, since a tightening arriving after house prices have been rising produces no additional cut. Equity is weak unconditionally but strongly state-dependent. At short horizons equity exposure cushions the response, as liquidity does, but when the tightening follows a quarter of strong share price growth the cushion is eroded and reversed, the effect turns significantly negative from the second year onward. The timing of the three channels reproduces the payoff-horizon ordering. Liquidity transmits on impact, housing over the first quarters, and equity only from the second year. The model delivers the same ranking the data do. The bond saver and the equity holder contract by less than the average household, while the hand-to-mouth agent and the collateral-constrained borrower contract by more.

This paper's contribution is twofold. First, it examines three balance-sheet channels, cash-flow, housing collateral, and equity valuation, within a common empirical and theoretical framework rather than separately, which allows their relative strength and timing to be compared directly. Second, and more importantly, the two approaches answer different questions. A portfolio regression identifies the sign of a channel but not the mechanism behind it. The liquidity differential is positive, but the regression cannot say whether liquid households are cushioned because they smooth or because they are on the receiving end of a transfer. The model separates the two, and reproduces the same positive liquidity sign with no consumption smoothing. The

bond holder is cushioned because the unexpected deflation raises the real value of its nominal claim and the borrower's deleveraging reaches it as loan repayment. Both transfers cancel in the aggregate, so the mechanism at play here is redistribution. This alone reproduces the empirical sign, which shows that the sign does not by itself identify the mechanism.

Though the model shows that redistribution alone can reproduce the liquidity sign, this remains a four agent economy in which each band holds a single asset. Real households hold diversified portfolios in which smoothing and redistribution operate together, so the model isolates a mechanism rather than measuring its share. On identification, the differential is estimated largely from between-country variation in portfolio composition, because consumption is published only as a national aggregate and is not available by decile. The within-country decile dimension therefore refines the exposure measure without adding variation in the outcome, and the estimates are best read as conservative and attenuated. The HFCS is rich cross-sectional microdata but a single wave at a fixed date, so portfolios are treated as time-invariant and the panel dimension the ECB has signalled but not yet released would be needed to let exposures move over time. In the model, the near-flat labour supply used to obtain correctly signed transmission compresses the response of hours and should be read with caution, though because labour productivity and elasticity are common across agents the ordering of the four types is unaffected.

These limitations point to a natural extension. A full heterogeneous-agent model would give realistic magnitudes driven by MPCs, and also let housing and equity affect consumption through the gains and losses households actually take on those assets, which the fixed stocks in this model rule out. Furthermore, with a higher-order solution it could also show the state dependence a linear model misses, which is the equity result we find in the data. The four agent model used here is a deliberate, tractable device for isolating mechanisms, and a HANK model is its quantitative successor. Whether portfolio composition shapes transmission is answered here, how much it shapes the magnitudes is the question that a HANK model would answer. Portfolio composition shapes how monetary policy reaches consumption, and it does so in an order the data reveal and a tractable model can reproduce. What remains is to measure how large these differences are.

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APPENDIX I COVID TREATMENT: YEAR FIXED EFFECTS

Table I.1 re-estimates the baseline liquidity specification under year fixed effects instead. γ_h keeps its sign and is, at every horizon, larger and more precisely estimated, it remains distinguishable from zero at the 90% level through $h = 6$, against $h = 3$ in the baseline. It is expected that the year dummies absorb the euro area-wide component of consumption, which is orthogonal to a differential identified off cross-sectional exposure, so γ_h gains precision without any change in its identifying variation. The aggregate response β_h, now identified from within-year variation, remains negative at every horizon and significant at the 90% level through $h = 7$. The liquidity channel is therefore robust to the way the pandemic is absorbed.

Table I.1: **Robustness: year fixed effects.** Liquidity specification.

	Covid dummy (baseline)				**Year fixed effects**			
	β_h		γ_h		β_h		γ_h	
h	Coef.	s.e.	Coef.	s.e.	Coef.	s.e.	Coef.	s.e.
0	−2.237*	1.114	0.116*	0.050	−2.214*	0.804	0.119*	0.045
1	−4.128*	2.501	0.177*	0.096	−2.965*	1.317	0.182*	0.083
2	−2.289*	0.899	0.113*	0.061	−2.211*	0.884	0.131*	0.057
3	−4.495*	2.087	0.140*	0.078	−4.325*	2.265	0.156*	0.074
4	−4.190*	1.878	0.098†	0.067	−2.766*	1.303	0.118*	0.059
5	−3.551*	1.401	0.109†	0.079	−2.303*	0.962	0.138*	0.066
6	−2.731*	1.334	0.101†	0.092	−3.070*	1.467	0.140*	0.080
7	−2.622*	1.377	0.032	0.085	−2.405*	0.873	0.074†	0.070
8	−1.361	1.609	0.008	0.106	−1.280†	0.963	0.068	0.087

Notes. Both blocks estimate Equation (1) under the baseline $8/6/4$ lag structure.* $|t| \geq 1.645$ (90%); † $|t| \geq 1$ (68%). Driscoll–Kraay SE.

APPENDIX II DECILE – VIGINTILE

Table II.1 re-estimates the baseline liquidity specification with the exposure aggregated over twenty within-country wealth bins rather than ten. The estimates are slightly smaller but preserve the same sign and significance profile at every horizon. This attenuation is expected when finer bins hold fewer households and measure cell-level exposure with more noise. The liquidity channel is therefore not an artefact of the chosen partition.

Table II.1: **Robustness: decile versus vigintile.** Liquidity-exposure differential γ_h.

	Decile (baseline)		**Vigintile**	
h	Coef.	s.e.	Coef.	s.e.
0	0.116^{*}	0.050	0.111^{*}	0.047
1	0.177^{*}	0.096	0.168^{*}	0.090
2	0.113^{*}	0.061	0.107^{*}	0.058
3	0.140^{*}	0.078	0.134^{*}	0.074
4	$0.098^{\dagger}$	0.067	$0.094^{\dagger}$	0.063
5	$0.109^{\dagger}$	0.079	$0.103^{\dagger}$	0.075
6	$0.101^{\dagger}$	0.092	$0.097^{\dagger}$	0.087
7	0.032	0.085	0.030	0.081
8	0.008	0.106	0.007	0.101

Notes. Both columns report estimates of the coefficient γ_h from Equation (1), with the liquidity index aggregated over within-country wealth deciles and vigintiles respectively, under the identical $8/6/4$ lag structure. * $|t| \geq 1.645$ (90%); † $|t| \geq 1$ (68%). Driscoll–Kraay SE.

APPENDIX III LAG-ORDER SELECTION

Table III.1 reports the information-criterion minima from a grid search at horizon $h = 0$, over $p_y \in \{1, \ldots, 8\}$, $p_X \in \{1, \ldots, 6\}$, and $p_s \in \{0, 2, 4\}$. In all three specifications both criteria select the same order, which should not be over interpreted because the selected orders sit at the upper boundary of the grid. The criteria point to ever-richer dynamics rather than settling on an interior optimum. We therefore treat the selection as indicative only, and support the lag-rich baseline with the robustness in Table III.2 and the lag-augmentation argument of Montiel Olea and Plagborg-Møller (2021) under which additional outcome lags restore valid inference at the cost of efficiency but not validity.

Table III.1: **Lag-order selection.** Across exposure specifications.

Specification	p_y	p_X	p_s	AIC	BIC	N
Liquidity	8	6	4	42476.2	42824.2	10400
Housing	8	6	4	42470.6	42818.6	10400
Equity	8	6	4	42476.4	42824.3	10400

Table III.2 presents the baseline liquidity specification under different lag selections. Extending the outcome lags past the grid (10/6/4) leaves the responses numerically indistinguishable from the baseline at every horizon, so pushing beyond the selected order does not move the estimates. The parsimonious 4/4/2 specification, which drops almost half of the baseline lag terms, leaves γ_h slightly larger at every horizon and significant at the 90% level one horizon further out. The baseline is thus the conservative choice, the differential is stable under both a large expansion and a large contraction of the lag structure, and is not an artefact of the chosen orders.

Table III.2: **Robustness: lag structure.** Liquidity differential γ_h.

	8/6/4		8/6/2		4/6/4		4/4/2		10/6/4	
h	γ_h	s.e.	γ_h	s.e.	γ_h	s.e.	γ_h	s.e.	γ_h	s.e.
0	0.116*	0.050	0.113*	0.051	0.126*	0.056	0.128*	0.060	0.116*	0.050
1	0.177*	0.096	0.166*	0.091	0.187*	0.102	0.179*	0.100	0.175*	0.096
2	0.113*	0.061	0.119*	0.063	0.128*	0.068	0.136*	0.073	0.110*	0.062
3	0.140*	0.078	0.150*	0.082	0.155*	0.084	0.164*	0.088	0.137*	0.078
4	0.098†	0.067	0.100†	0.067	0.112*	0.059	0.111*	0.064	0.094†	0.066
5	0.109†	0.079	0.110†	0.078	0.125*	0.075	0.118†	0.079	0.106†	0.078
6	0.101†	0.092	0.107†	0.094	0.121†	0.092	0.124†	0.101	0.097†	0.090
7	0.032	0.085	0.046	0.085	0.049	0.079	0.056	0.083	0.027	0.084
8	0.008	0.106	0.025	0.104	0.028	0.102	0.030	0.106	0.004	0.105

Notes. Each column re-estimates Equation (1) under the lag structure $p_y/p_X/p_s$ shown in the header, with everything else as in the baseline. * $|t| \geq 1.645$ (90%); † $|t| \geq 1$ (68%). Driscoll–Kraay SE.

We re-estimate θ_h under the 4/4/2 structure. Because the state-dependent response emerges only at $h \geq 5$, the pertinent check contracts the lag structure rather than extending it. Table III.3 shows the negative θ_h at medium-to-long horizons is preserved and, if anything, larger in magnitude with significance emerging at the same horizon and at the same level. The valuation channel is therefore not an artefact of the lag-rich baseline. The housing specification is not shown as its θ_h is insignificant at every horizon and offers no result to overturn.

Table III.3: **Robustness: lag structure, equity.** State-dependent coefficient θ_h under a parsimonious lag structure.

	8/6/4 (baseline)		4/4/2	
h	θ_h	s.e.	θ_h	s.e.
0	−0.157	0.461	−0.060	0.513
1	0.139	0.827	0.175	0.844
2	−0.439	0.475	−0.377	0.518
3	−0.395	0.819	−0.460	0.789
4	−0.494	0.871	−0.563	0.880
5	−1.322*	0.747	−1.509*	0.805
6	−1.578*	0.803	−1.794*	0.839
7	−2.205*	0.808	−2.560*	0.859
8	−2.232*	0.846	−2.546*	0.872

Notes. Both columns estimate Equation (3) with the equity share and share price terms, under the lag structure $p_y/p_X/p_s$ shown. * $|t| \geq 1.645$ (90%); † $|t| \geq 1$ (68%). Driscoll–Kraay SE.

APPENDIX IV STATIONARITY TESTS

IV.1 Stationarity of the Input Series

Table IV.1 reports ADF, PP and KPSS tests on every series entering the projections. ADF and PP test for a unit root, KPSS for level stationarity, and a series is classified stationary only when the two unit-root tests reject and KPSS does not.

House price growth is the exception. Under the baseline four-quarter measure neither unit-root test rejects in eight out of ten countries; under a one-quarter difference the PP test rejects in nine of ten. The four-quarter window is an overlapping difference that mechanically inflates measured persistence, so this is a property of the transformation rather than the underlying series. The samples are also short so the tests have limited power.

We keep the year-on-year measure as the baseline. The ECB index is not seasonally adjusted, so a four-quarter difference removes seasonality by construction, whereas a one-quarter difference needs a separate correction (Appendix IV.2). Persistence does not threaten the result. House price growth enters Equation (2) only through the control ρ_h, and the state interaction θ_h, the latter insignificant at every horizon under both measures. Since consumption growth and the shock are stationary, the differential γ_h, identified from the shock interaction, is unaffected by how persistent the control is.

Table IV.1: **Stationarity tests.** All input series, by country.

Country	ADF	PP	KPSS	Tests for stationarity
Panel A: Monetary policy shock, QoQ consumption growth, QoQ GDP growth, QoQ share price return: all 3/3, not listed individually				
Panel B: QoQ HICP inflation — 7/10 pass all three				
AT	≤ 0.01	≤ 0.01	0.024	2/3
DE	0.058	≤ 0.01	0.068	2/3
IE	0.074	≤ 0.01	≥ 0.10	2/3
Panel C: Change in the unemployment rate — 6/10 pass all three				
ES	0.163	≤ 0.01	≥ 0.10	2/3
IE	0.161	≤ 0.01	≥ 0.10	2/3
NL	0.090	≤ 0.01	≥ 0.10	2/3
PT	0.053	≤ 0.01	≥ 0.10	2/3
Panel D: YoY house price growth (baseline) — 0/10 pass all three				
AT	0.018	0.381	≥ 0.10	2/3
BE	0.078	0.531	≥ 0.10	1/3
DE	0.318	0.338	≥ 0.10	1/3
ES	0.223	0.518	≤ 0.01	0/3
FI	0.171	0.074	≤ 0.01	0/3
FR	0.202	0.311	≥ 0.10	1/3
IE	0.236	0.629	0.032	0/3
IT	≤ 0.01	0.392	≤ 0.01	1/3
NL	0.567	0.349	≤ 0.01	0/3
PT	0.347	0.428	≤ 0.01	0/3
Panel E: QoQ house price growth — 1/10 pass all three (FR)				
AT	0.399	≤ 0.01	≥ 0.10	2/3
BE	0.155	≤ 0.01	≥ 0.10	2/3
DE	0.242	≤ 0.01	≥ 0.10	2/3
ES	0.083	0.015	0.021	1/3
FI	0.045	≤ 0.01	≤ 0.01	2/3
IE	0.072	0.326	0.047	0/3
IT	0.272	≤ 0.01	≤ 0.01	1/3
NL	0.057	≤ 0.01	≤ 0.01	1/3
PT	0.291	≤ 0.01	≤ 0.01	1/3

Notes. Reported values are p-values, with bounds where the statistic falls outside the test's tabulated range. The last column counts how many of the three tests point to stationarity: ADF or PP with $p < 0.05$ (rejecting a unit root), and KPSS with $p > 0.05$ (not rejecting stationarity). A series passes when all three agree (3/3); only countries below 3/3 are listed, all others pass.

IV.2 Quarter-on-Quarter Housing Price

Table IV.2 reports Equation (2) re-estimated with $\widetilde{HP}_{c,t-1}$ built from quarterly rather than annual house price growth. The quarterly rate carries a seasonal pattern (the index is not seasonally adjusted), significant in five countries and removed here as country-specific calendar-quarter means. The need for this correction, which the year-on-year measure avoids by construction, is why the latter is preferred in the baseline. The exposure differential is negative over the first horizons under both measures and slightly stronger under the quarterly one, reaching 90% on impact. θ_h reaches only the 68% band at short horizons under the quarterly measure and is significant at no horizon in either, so the absence of a housing-market state effect does not depend on how the state is measured. The weakly positive γ_h at $h = 7$–8 in the baseline disappears here, so the robust content of the housing result is the negative short-horizon differential, which both measures share.

Table IV.2: **Robustness: housing state variable.** Year-on-year versus quarter-on-quarter house price growth.

	YoY (baseline)				**QoQ, seasonally demeaned**			
	γ_h		θ_h		γ_h		θ_h	
h	Coef.	s.e.	Coef.	s.e.	Coef.	s.e.	Coef.	s.e.
0	−0.055†	0.035	0.020	0.165	−0.091*	0.038	0.008	0.069
1	−0.149†	0.106	0.049	0.290	−0.149†	0.119	0.293†	0.186
2	−0.100	0.107	0.111	0.220	−0.137†	0.108	0.177†	0.120
3	−0.033	0.098	0.008	0.295	−0.082	0.090	0.190†	0.169
4	0.059	0.094	−0.074	0.305	−0.017	0.091	0.113	0.198
5	0.109†	0.094	−0.017	0.313	0.029	0.104	0.124	0.212
6	0.076	0.108	−0.222	0.363	−0.038	0.101	−0.045	0.211
7	0.172†	0.112	−0.115	0.333	0.057	0.123	−0.014	0.226
8	0.204†	0.139	0.029	0.369	0.087	0.137	0.037	0.231

Notes. Both blocks estimate Equation (2) under the baseline 8/6/4 lag structure, differing only in how $\widetilde{HP}_{c,t-1}$ is built. * $|t| \geq 1.645$ (90%); † $|t| \geq 1$ (68%). Driscoll–Kraay SE. γ_h is comparable across blocks. θ_h is not, the state variable being scaled by 10 pp in the baseline and by one standard deviation on the right.

APPENDIX V CALIBRATION ROBUSTNESS

Table V.1 varies the inverse Frisch elasticity φ and the loan-to-value ratio m_H, and tracks two signs of correct transmission. The first is the impact response of the nominal policy rate di_0, which a contractionary shock should raise. At $\varphi = 1$ and at $\varphi = 0.33$ it is negative, the Taylor-rule response to falling inflation overturns the direct tightening and the rate falls on impact. It turns positive only at the baseline $\varphi = 0.01$, or at $\varphi = 0.33$ once m_H is lowered to 0.08. The second is the collateral multiplier's volatility relative to its steady-state value, $\sigma_\mu/\bar{\mu}$, which measures how close the always-binding constraint comes to failing. At $\varphi = 1$ it falls from 1.55 to 0.05 at baseline, so a constraint that is nearly slack at high φ becomes firmly binding. Both improve as φ falls, and a lower loan-to-value ratio works in the same direction.

Table V.1: **Sign of monetary transmission.** Across labour-supply elasticities.

	φ	m_H	di_0	$\sigma_\mu/\bar{\mu}$
Unit Frisch elasticity	1.00	0.1229	−0.431	1.55
Intermediate	0.33	0.1229	−0.065	0.87
Intermediate, reduced LTV	0.33	0.0800	+0.116	0.61
Baseline	0.01	0.1229	**+0.489**	**0.05**

Notes. di_0 is the impact response of the nominal policy rate to a contractionary monetary shock. $\sigma_\mu/\bar{\mu}$ is the standard deviation of the collateral multiplier over its steady-state value. Responses are first-order decision-rule coefficients on ε^v in model units. Each row is re-calibrated so that $N_i = 1/3$ and $q_h\bar{H}/(4Y) = 3.559$ continue to hold.

The baseline $\varphi = 0.01$ is not a literal labour-supply elasticity but a reduced-form device, as set out in Section 7. The table reproduces textbook transmission only through it, which we treat as a limitation rather than a result: what the exercise establishes is the sign and ordering across rows, not the precise magnitudes.

APPENDIX VI CODE AND REPRODUCIBILITY

The code reproduces both parts of the thesis. The empirical part estimates local projections on a stacked country × wealth decile × quarter panel using Jarociński–Karadi high-frequency monetary policy surprises in R; the structural part is a four agent New Keynesian model calibrated to German HFCS data and solved in Dynare.

DATA AND CONFIDENTIALITY. The HFCS microdata (Wave 4, User Database 4.0, ASCII) is confidential and cannot be redistributed. It is therefore excluded from the deposited code. End-to-end reproduction requires the reader's own applicant copy, obtained through the ECB, with its location set in the single `base_path` line of `0.config.R`. The seven public macro series (ECB Data Portal, FRED/OECD, and the Jarociński–Karadi surprises) are stored with the code as frozen snapshots. Without the microdata, scripts `3.1`–`3.3` still reproduce the local-projection estimates from the stored country-by-decile exposure files, and `5.tank_results.R` reproduces the model figures from the committed `irfs_baseline.csv` without data or MATLAB; only scripts `1` to `2.4` and the descriptive step in `4.tables_graphs.R` require the household files.

SOFTWARE. The empirical pipeline runs on R 4.6.0 (Windows 11), using `data.table`, `survey`, `mitools`, `lubridate`, `sandwich`, `ggplot2`, `readxl` and `tseries`. The four agent TANK model is solved in Dynare 6.4 under MATLAB R2024b.

PIPELINE. `run_all.R` sources the ten scripts in order and reports `ok` or `FAIL` for each, a full run takes about five minutes. All specification choices sit in the `cfg_*` list at the head of each script. Scripts `3.1`–`3.3` produce the local-projection results reported in the empirical section. Script `5` produces the model impulse responses of the four agent model in the results section. Table VI.1 lists each script.

MODEL. `dynare/tank_4agents_of.mod` is solved by hand in Dynare, and `export_irfs.m` writes `irfs_baseline.csv`:

```
cd dynare
dynare tank_4agents_of.mod
export_irfs
```

Because `irfs_baseline.csv` is in the deposited code, `5.tank_results.R` reproduces the

model figures without MATLAB. Calibration targets come from `check_calibration_targets.R`.

Table VI.1: Reproduction pipeline (`run_all.R`) and auxiliary scripts.

Script	Task	Writes
`1.load_data.R`	reads and merges the HFCS ASCII files per implicate; keeps the ten countries and analysis variables	`hfcs_light.RData`
`2.1.distributions_bins.R`	within-country wealth deciles and vigintiles	`hfcs_light_enriched.RData`
`2.2.liquidity_index.R`	exposure 1: graded liquidity index	`exposure_liquidity_index_*`
`2.3.housing_share.R`	exposure 2: housing asset share	`exposure_housing_asset_share_*`
`2.4.equity_exposure.R`	exposure 3: equity share	`exposure_equity_share_*`
`3.1.LP_liquidity_index.R`	local projections, liquidity interaction	`lp_results_liquidity_864.csv`
`3.2.LP_house.R`	local projections, housing; house price triple interaction	`lp_results_housing_864.csv`
`3.3.LP_equity.R`	local projections, equity; share price triple interaction	`lp_results_equity_864.csv`
`4.tables_graphs.R`	descriptive tables and figures; all LP figures	`table_4_{1,2}.csv`, figures
`5.tank_results.R`	model impulse responses	`fig_irf_C_{level,cum,diff}.pdf`, `fig_irf_macro.pdf`, `tab_irf_results.csv`
Outside `run_all.R` — one-offs and robustness checks		
`check_calibration_targets.R`	steady-state targets pasted into the `.mod`	`calib_*.csv`
`check_lag_selection.R`	AIC/BIC grid behind the 8/6/4 lag choice	`aic_bic_*.csv`
`check_stationarity.R`	ADF, PP and KPSS on every LP input	`stationarity_tests.csv`
`RC.LP_vigintile_liquidity.R`	robustness: twenty wealth bins instead of ten	`lp_results_*.csv`
`RC.LP_house_qoq.R`	robustness: quarter-on-quarter housing state	`lp_results_*.csv`

Notes. Numeric prefixes give run order; `run_all.R` sources scripts 1–5 in sequence. Files under `data/processed/` are intermediates, rebuilt on each run. Scripts 1 and 2.1 write household-level intermediates (`*.RData`) that are *not* distributed; all other processed files are cell-level aggregates. The five scripts below the rule sit outside `run_all.R` and feed no downstream step: three record settled choices, two are robustness checks.